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6. Creating a CentOS 8 Server Virtual Machine

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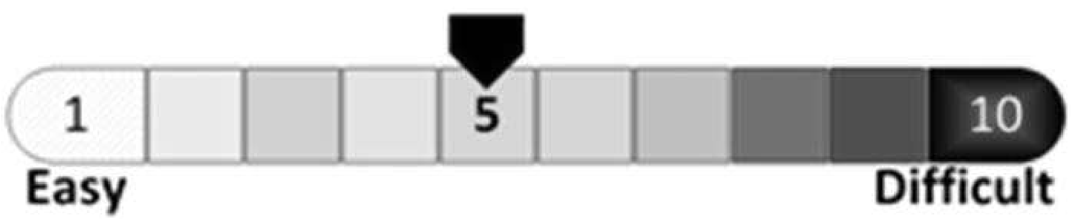
In this chapter, you will download a CentOS 8 bootable image to install and create a second virtual machine on your VMware Workstation. The CentOS 8 server is based on a community version of Red Hat Enterprise Linux 8 (RHEL8) , one of the most popular enterprise Linux OSs among the Linux community users because of its stability and security. By the end of this chapter, you will be able to create a CentOS 8 Server VM from scratch and customize the network settings using the `nmtui` package to add flexibility to your network automation lab. In preparation, you will download the GNS3 `.ova` template file and import and create a GNS3 VM for Cisco IOS and IOSv image integrations later.



In the previous chapter, you created an Ubuntu server machine and learned how to customize its settings to make it more usable for our lab. This chapter is one of the shorter chapters in this book, and in it you will create another Linux virtual machine using a CentOS 8 bootable image. Although we could simply clone the existing Linux VM to create more virtual machines, this prevents us from learning the basic operations of other Linux distributions. Therefore, we will download and install a second and different Linux distribution. You will learn how to turn this server into a server with multiple IP services for lab testing purposes in Chapter **8**. As in the previous process, you will search for and download the bootable image and then install the image and create a virtual machine. After creating the CentOS 8 virtual machine, you will test the server's connection via SSH and learn an easy way to manage the network adapter on a Linux server. To prepare for GNS3 integration, we will need to download the correct copy of GNS3 VM for VMware Workstation, so using the downloaded GNS3 VM `.ova` file, the GNS3 VM will be created ahead of time for smoother integration. Let's quickly create your second virtual machine.

Downloading and Installing a CentOS 8 Server Image

The actual process of downloading and installing a CentOS VM is identical to the Ubuntu 20.04 server installation. As in the Ubuntu 20.04 LTS VM installation, you must first download a bootable CentOS 8 installation image (`.iso`) file. First, check your Internet connection and follow the steps in this chapter to create and install your new CentOS 8 VM on VMware Workstation 15 Pro. At the time of writing this book, the latest CentOS version was 8.1.1804. By the time you read this book, newer versions will be available, but as long as you stick with the main 8.x (or 7.x) version, you will be able to follow along and successfully install your CentOS server illustrated in this book. If you opt to download and install another CentOS version, the installation steps will still be almost identical. In Chapter **8**, you will be installing different software to turn this server into a multipurpose lab server , so you may have to follow the software installation procedures suggested for your version of CentOS system because one version of software compatible on this CentOS version may not be compatible on previous or newer software versions. This CentOS VM also can be used as a multipurpose lab server running several IP services for any future IT certification preparations.

> ⚠ As IBM has purchased Red Hat, there have been some changes for future developments and support for CentOS servers. The community support for CentOS 8 servers will cease on December 31, 2021, and only CentOS 8 Stream or a newer Stream versions of CentOS will be available to community users. CentOS Stream is positioned between Fedora (the beta version of Red Hat Enterprise Linux) and Red Hat Enterprise Linux, so we can say it is the early release version of Red Hat Enterprise Linux. If you use Stream in production, there will be issues getting support from Red Hat (IBM), as this removes Red Hat's obligation to support a bug or problems in no longer supported CentOS server versions. However, using a Stream version in a lab environment is still acceptable as most functions will meet our testing needs.

First, let's download the latest CentOS 8 installation ISO file from the official CentOS site.

Downloading the CentOS 8 Server Image

#	Task
01	From your favorite web browser, go to the CentOS's official site suggested and click the x86_64 link on the web page. Optionally, you can choose to download the CentOS Stream version . See Figure **6-1**. URL: **`https://www.centos.org/download/`**

#	Task

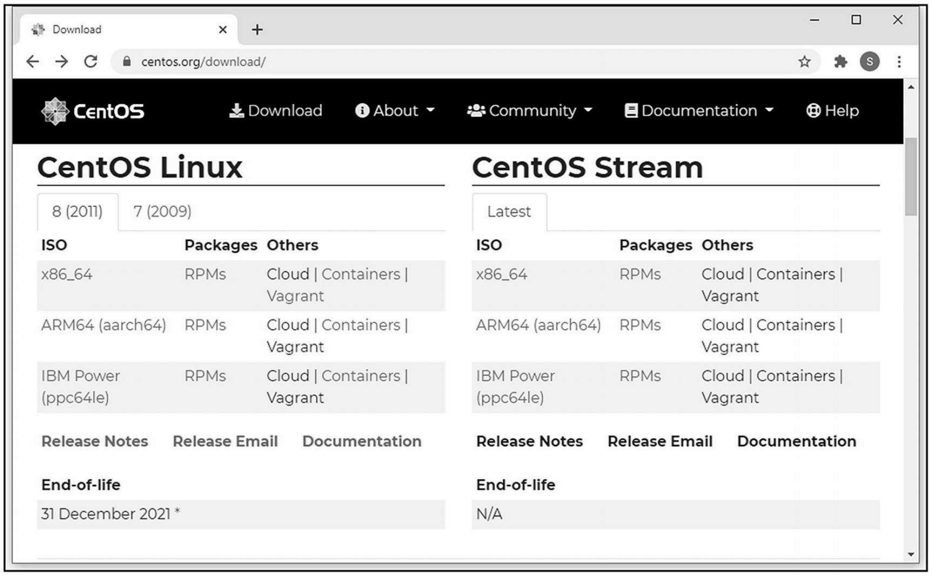


Figure 6-1. CentOS official download page

02 Next, locate the closest CentOS image download server to your location and download the latest CentOS 8 ISO from the FTP server. The estimated file size will vary depending on which file you download, but here you will download the file ending with `–x86-64-dvd1.iso`, which is more than 7GB in size. At the time of writing this book, the latest CentOS 8 file available was `CentOS-8.1.1911-x86_64-dvd1.iso`. See Figure **6-2**.

URL: **http://isoredirect.centos.org/centos/8/isos/x86_64/**

Task

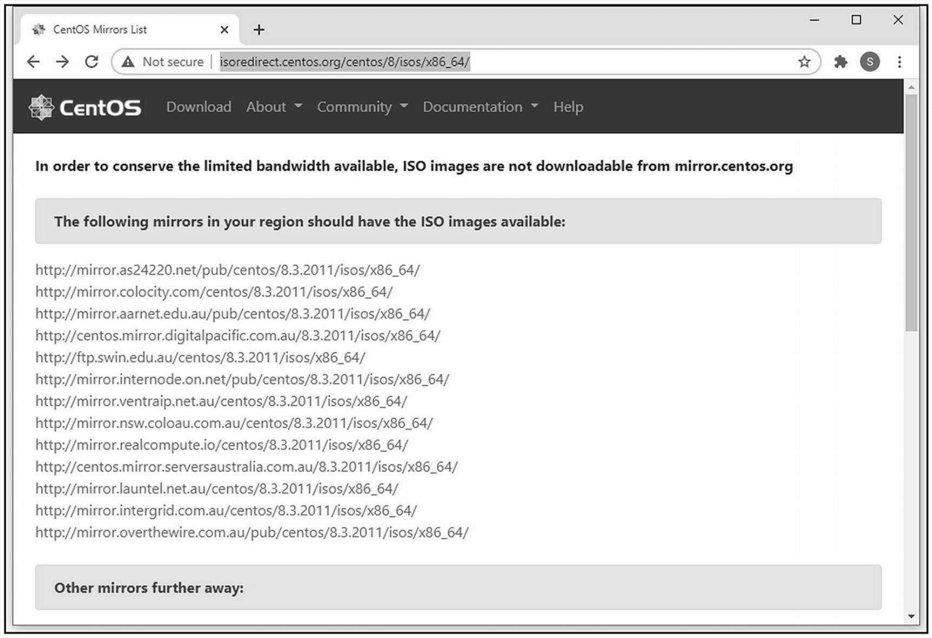


Figure 6-2. CentOS 8, official download page mirror example

After the software download completes and is saved to your `Downloads` folder, you are ready to create a CentOS VM from VMware Workstation 15 Pro.

Installing CentOS 8 Server

The CentOS 8 server VM installation process is like the Ubuntu server installation process, so let’s quickly go through this process and create a new server VM.

 # Task

01 In your VMware Workstation menu, select File ➤ New Virtual Machine . See Figure **6-3**.

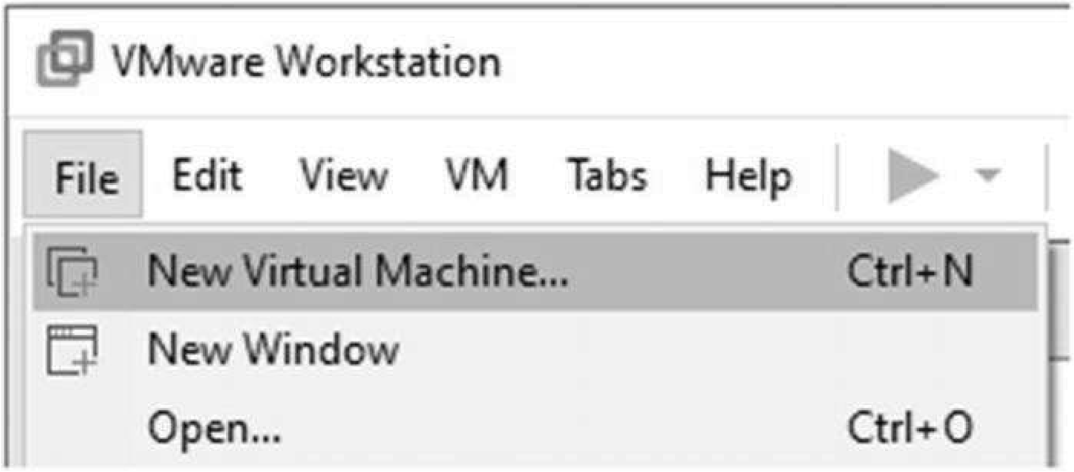


Figure 6-3. CentOS 8 installation, creating a new VM

02 When the New Virtual Machine Wizard appears , click the "Typical (recommended)" option and click the Next button. See Figure **6-4**.

Figure 6-4. CentOS 8 installation, New Virtual Machine Wizard start page

03 When you come back to the New Virtual Machine Wizard window shown in Figure 6-5, choose the “I will install the operating system later” option . If you choose the “Installer disc image file (iso)” option here, the CentOS 8 image will return `Pane is a dead error`, and you won’t be able to install the operating system. See Figure 6-5.

> **Note**Make sure you select “I will install the operating system later” here.

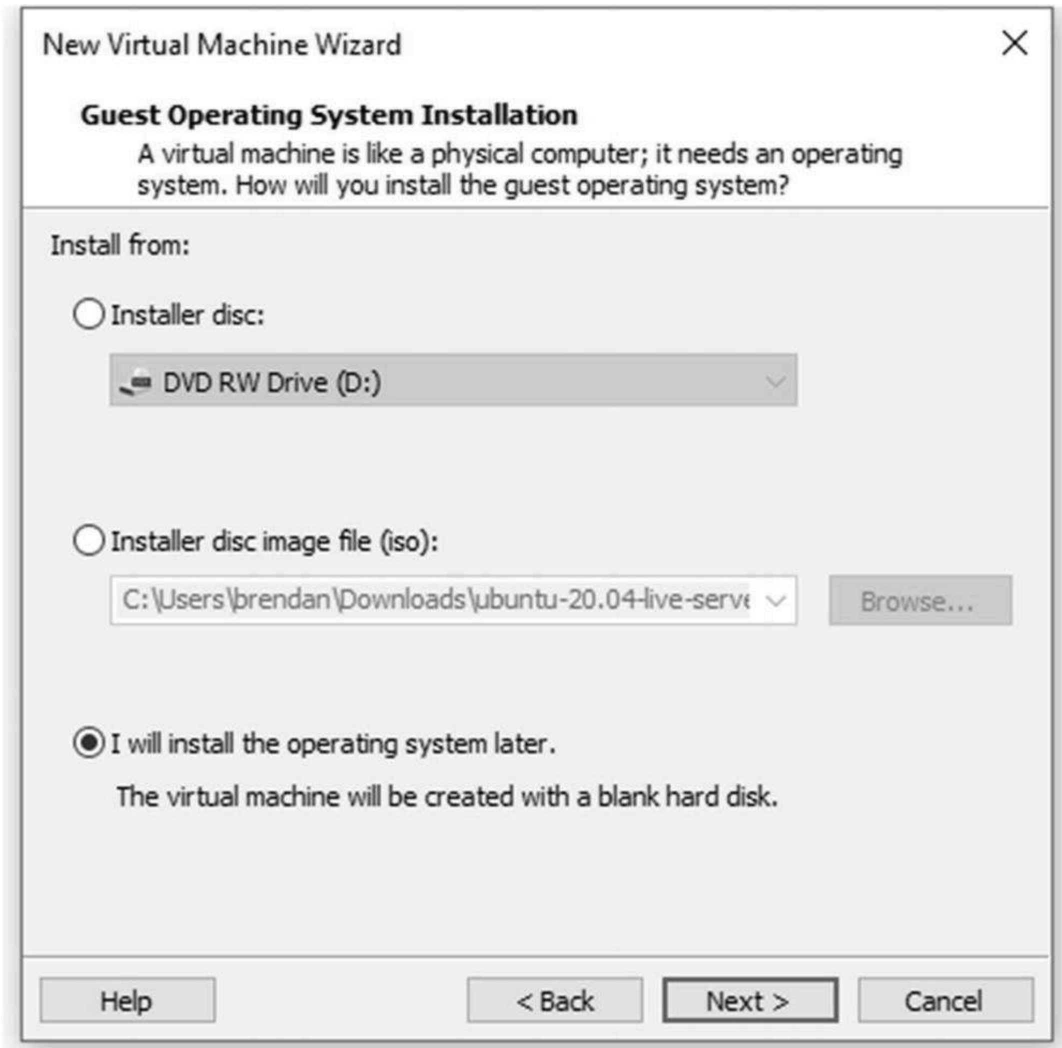


Figure 6-5. CentOS 8 installation, selecting the “I will install the operating system later” option

04 Depending on the version of VMware Workstation 15 you have installed on your host PC, the CentOS 8.x version may not be available. If you encounter this problem, you can always choose the Red Hat Enterprise Linux (REHL) 8 64-bit version of your choice as CentOS 8 is

the same version as RHEL 8 without the logo and proprietary contract. See Figure **6-6**.

NoteNote that IBM acquired Red Hat in July 2019. CentOS has been the community version of the current production Red Hat Enterprise Linux; when RHEL runs into a bug, Red Hat provided a fix for CentOS within three days. This made CentOS dependable as the RHEL Linux, but the users could still use CentOS in production. By contrast, CentOS Stream is a development preview or experimental version of RHEL's pre-release, and Fedora is ahead of CentOS Stream and even more experimental. On December 9, 2020, Red Hat (IBM) announced that CentOS Linux 8 will be unsupported at the end of 2021. This does not affect your lab environment; however, avoid deploying CentOS 8 in production. For your lab build, we can use CentOS Stream or Fedora to replace CentOS 8 after 2021.

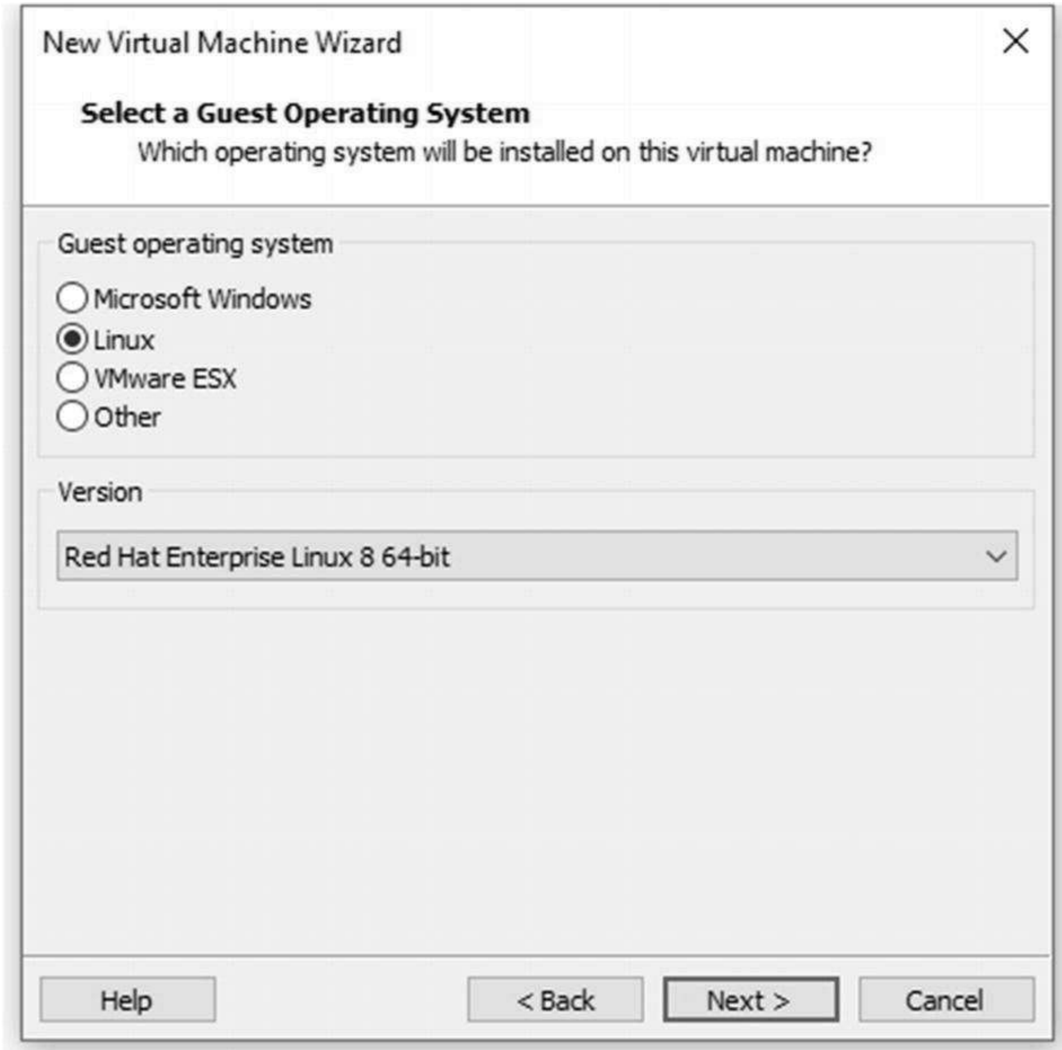


Figure 6-6. CentOS 8 installation, OS selection

05 In the next window, enter the user credentials as requested and click Next. See Figure 6-7.

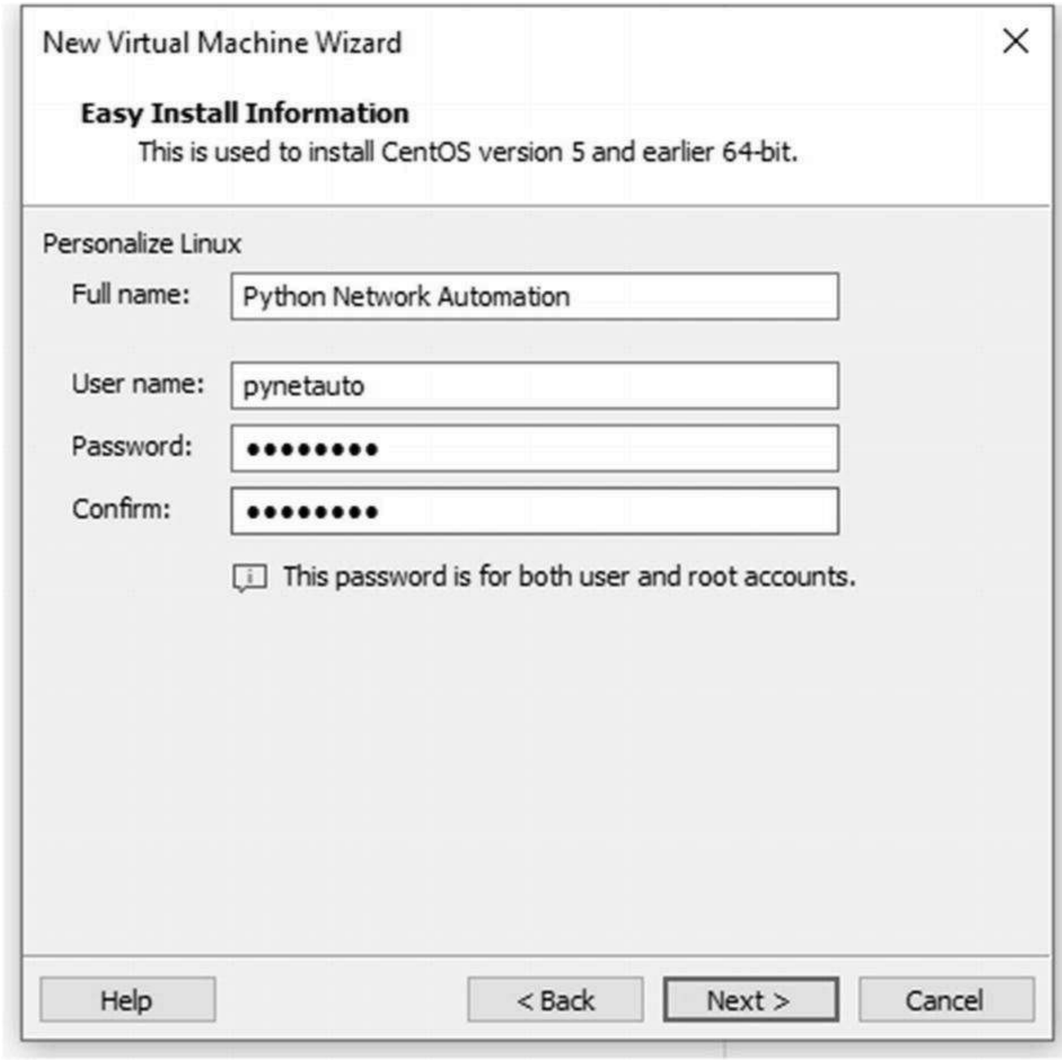


Figure 6-7. CentOS 8 installation, entering user credentials

06 Use some imagination to give your server a meaningful name that is easy to recognize. The server was named CentOS8s1, referring to CentOS8 Server 1. Click the Next button. See Figure **6-8**.

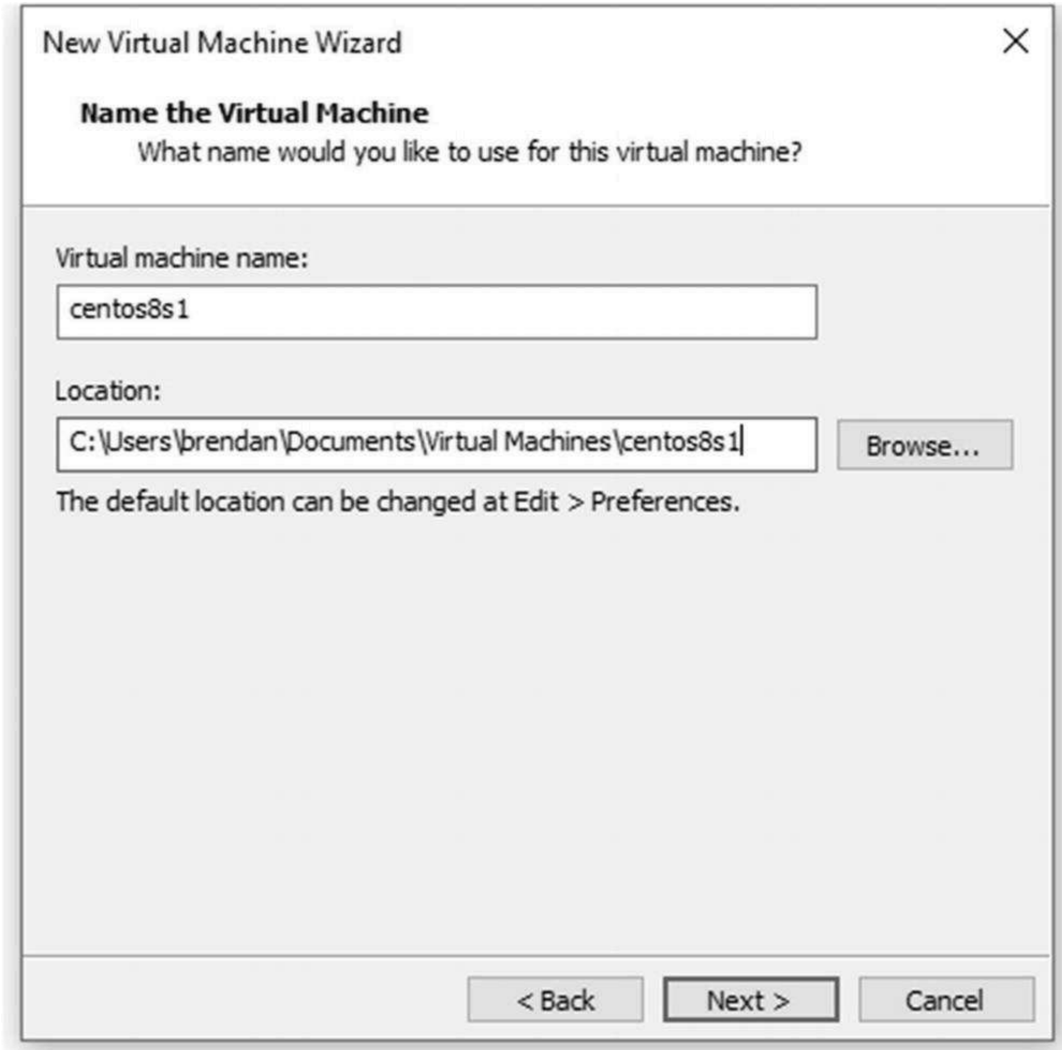


Figure 6-8. CentOS 8 installation, server name and folder configuration

07 Allocate the desired free disk space ; in the following example, 30GB disk space was allocated to the GUI, and other software will be installed. By default, the disk is set to thin provisioning, increasing disk size incrementally until it reaches 30GB so that the initial disk space will be a lot less. Also, click "Store virtual disk as a single file" for easier file management later. Then click Next. See Figure **6-9**.

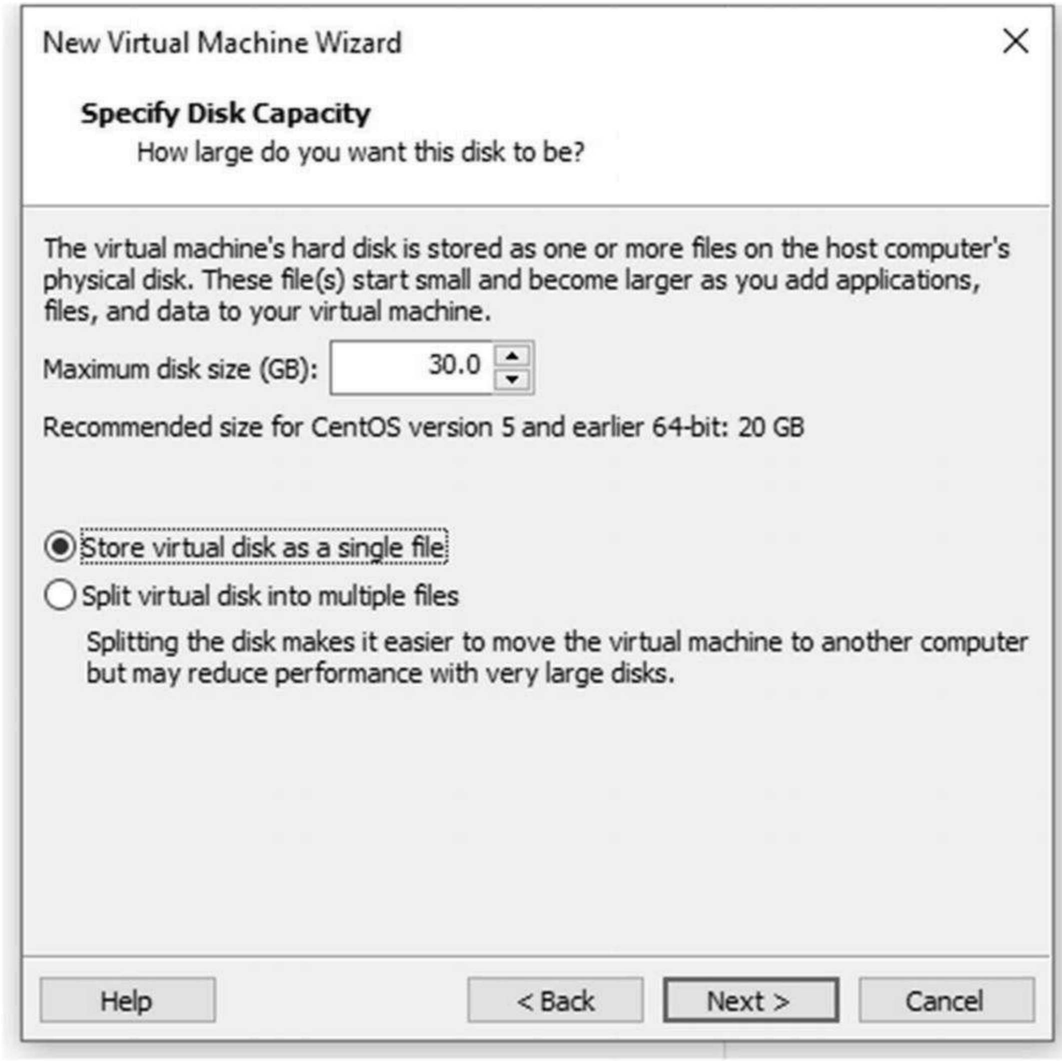


Figure 6-9. CentOS 8 installation, allocating disk capacity for use

08 Click the Finish button to finish the VM configuration . See Figure **6-10**.

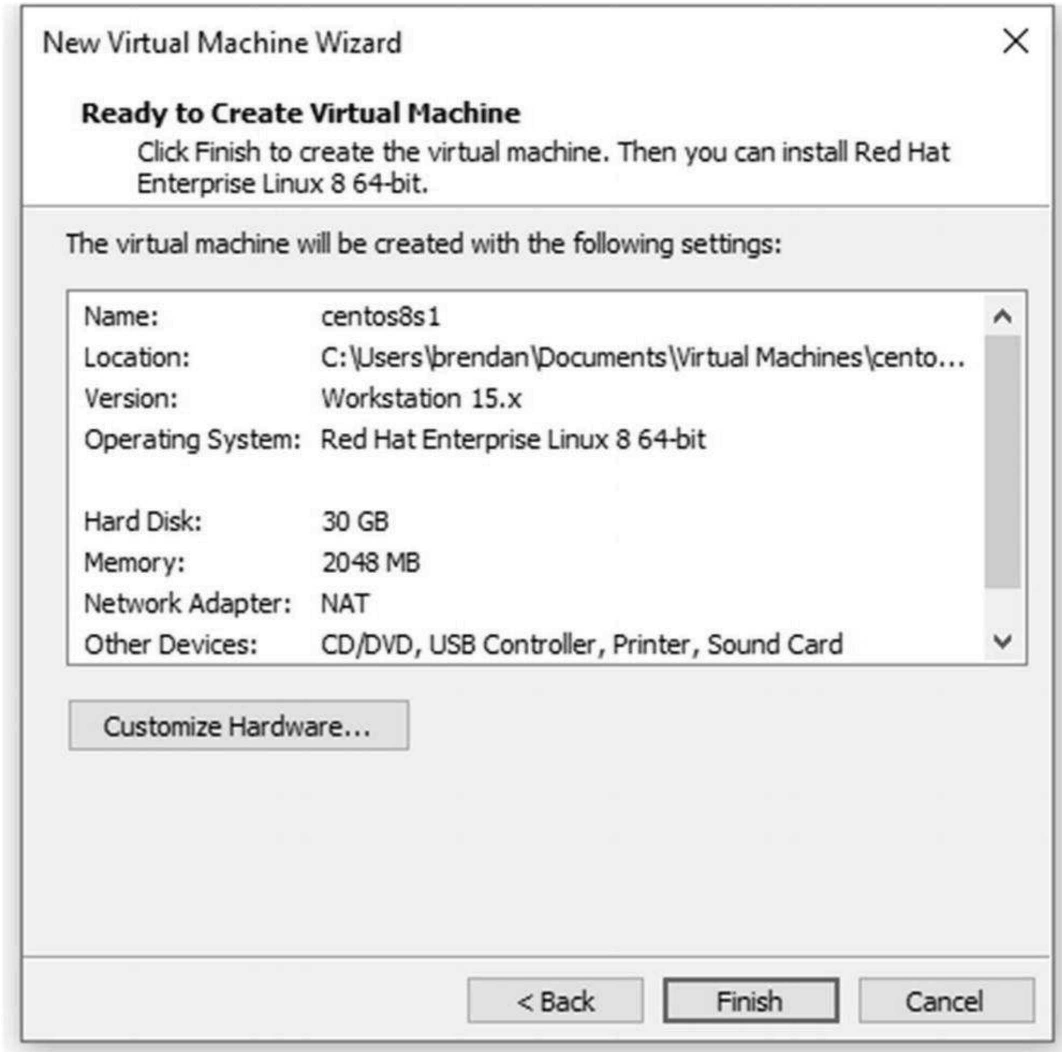


Figure 6-10. CentOS 8 installation, finishing the configuration

09 Now click “Edit virtual machine settings ” to change the CD/DVD selection. See Figure **6-11**.

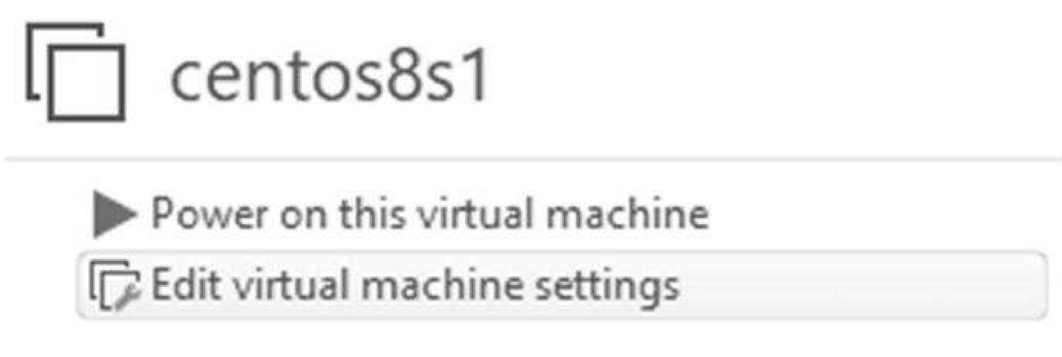


Figure 6-11. CentOS 8 installation, selecting “Edit virtual machine settings ”

10 Click CD/DVD (SATA) in the left pane, click “Use ISO image file,” and then click the Browse button. See Figure **6-12**.

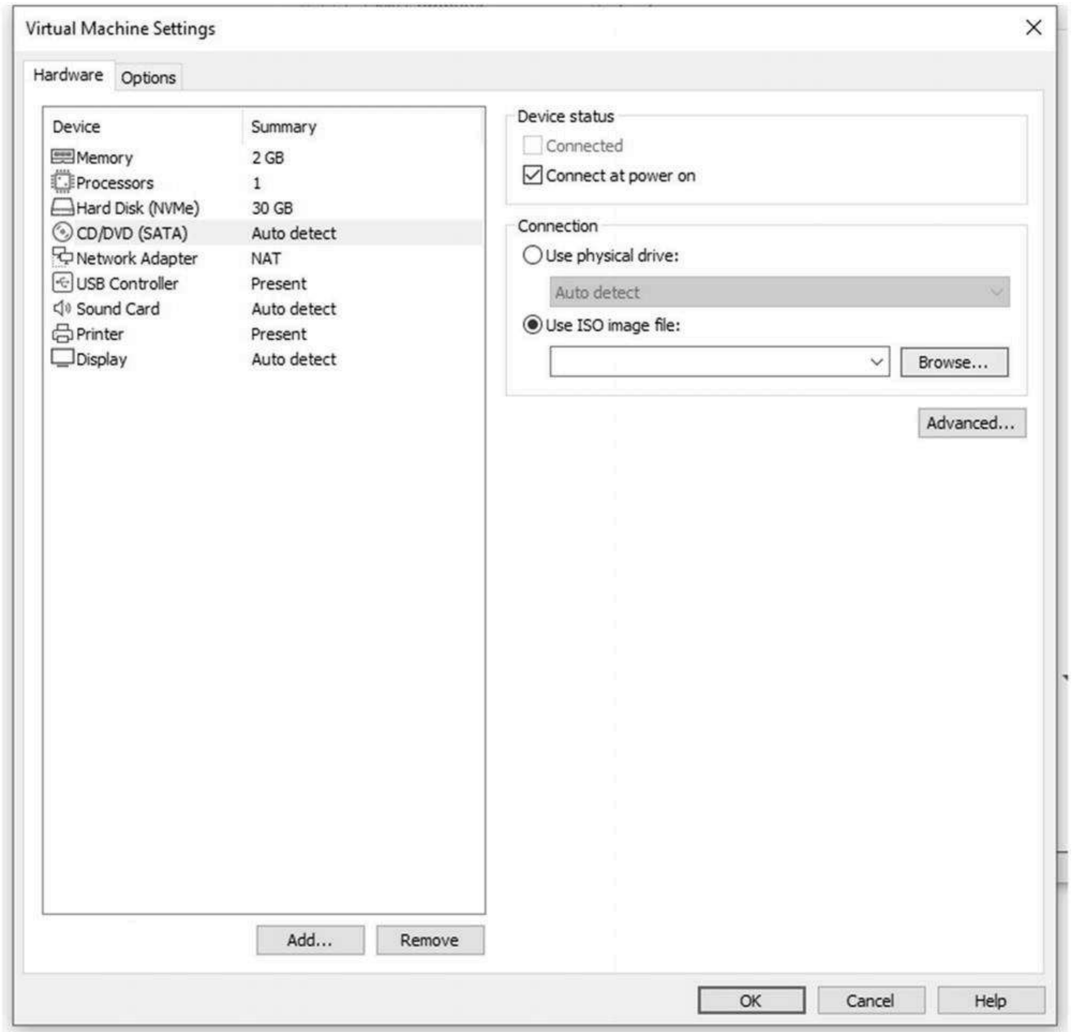


Figure 6-12. CentOS 8 installation, selecting virtual machine settings

11 Select the CentOS installation file already downloaded in your `Downloads` folder, and then click Open. See Figure **6-13**.

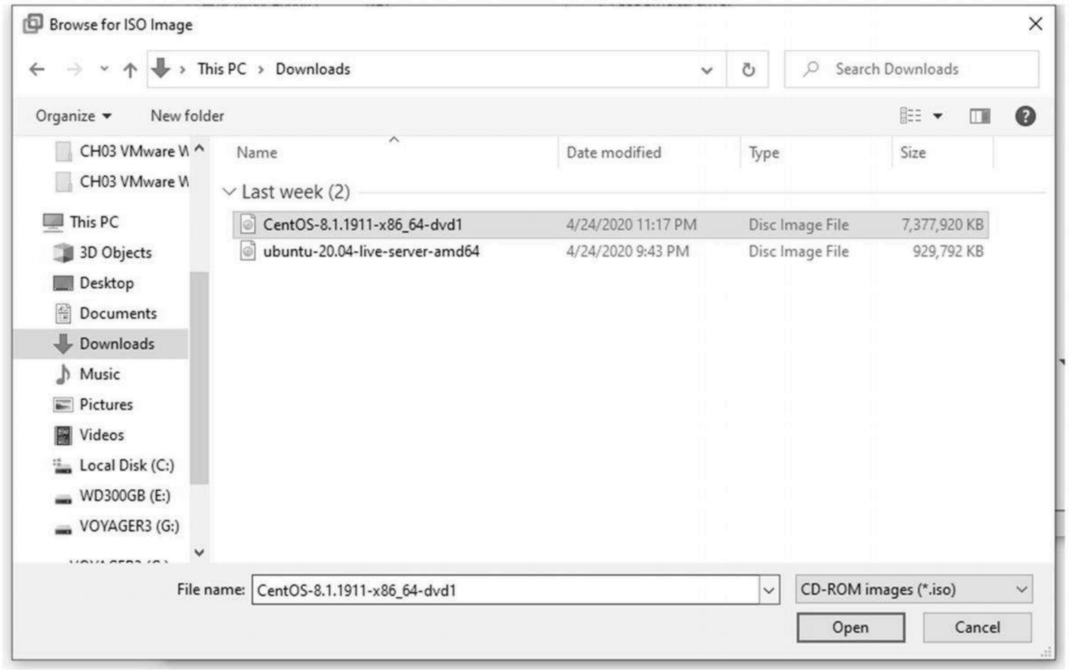


Figure 6-13. CentOS 8 installation, selecting the CentOS ISO file

12 Now click “Power on this virtual machine ” to start the CentOS installation. See Figure **6-14**.

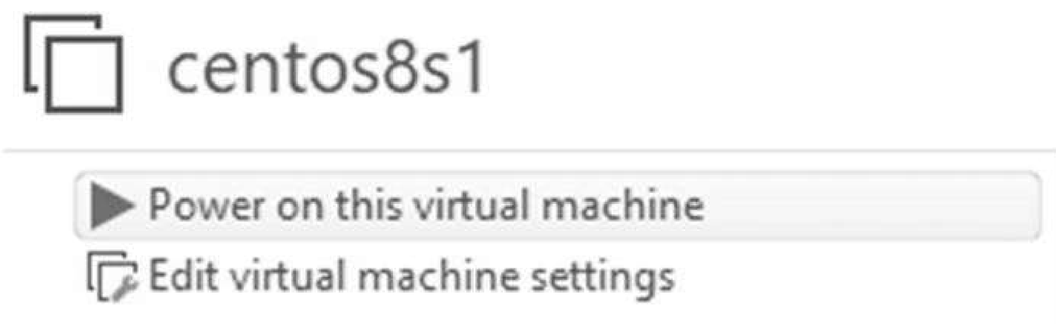


Figure 6-14. CentOS 8 installation, powering on hte VM

13 Click Install CentOS Linux 8 to begin the installation . See Figure **6-15**.

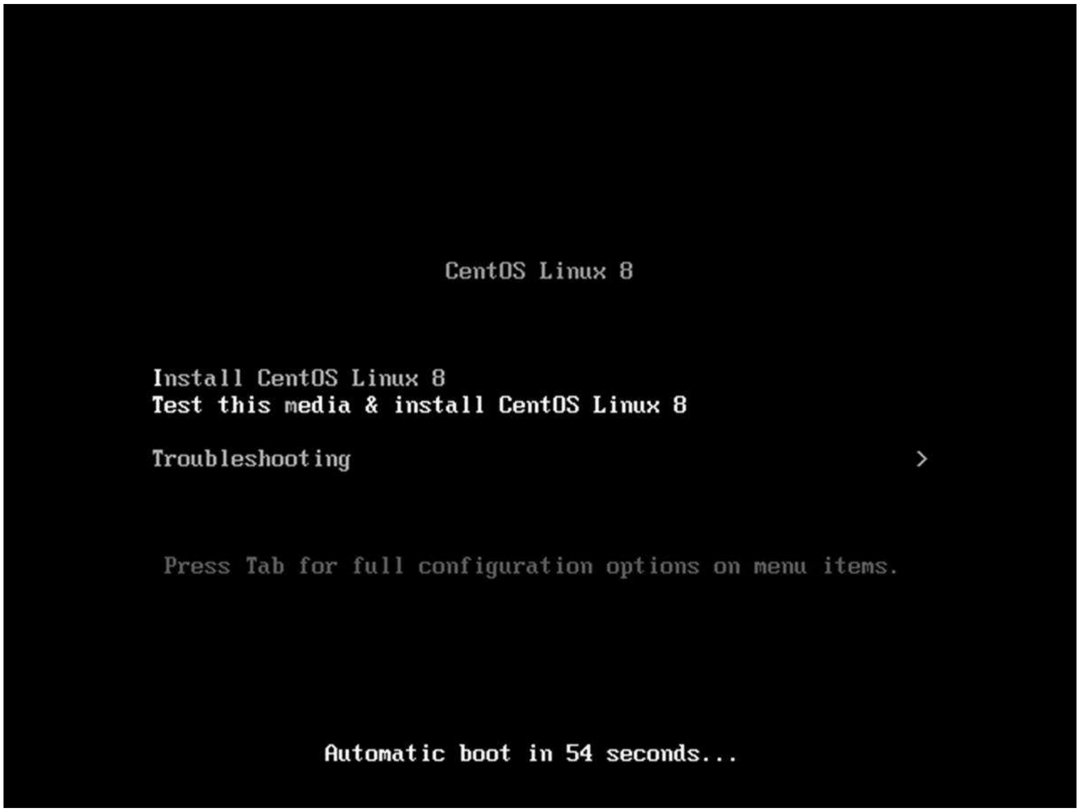


Figure 6-15. CentOS 8 installation, selecting installation options

14 Leave the default language as English (United States) and click Continue to move to the next page. See Figure **6-16**.

Figure 6-16. CentOS 8 installation, language selection

15 In the Installation Summary window , select Installation Destination under System to enter and select the operating system installation disk. See Figure 6-17.

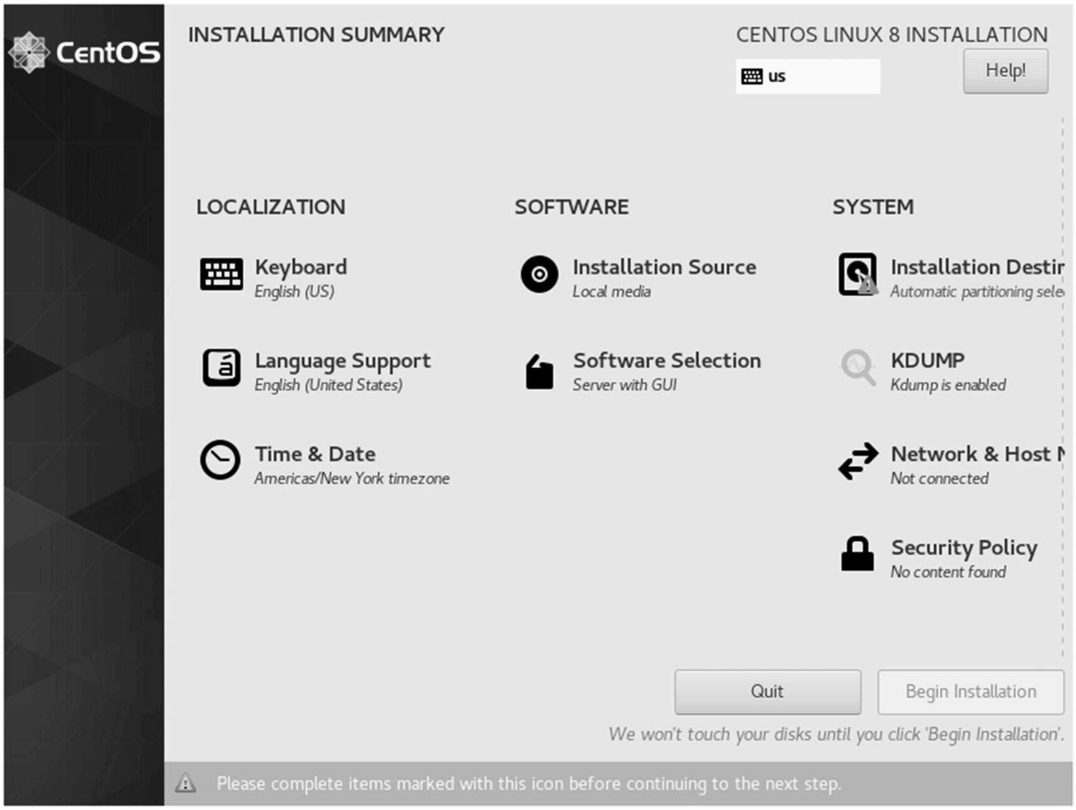


Figure 6-17. CentOS 8 installation, installation summary

16 Highlight VMware Virtual NVMe Disk 30 GiB and click the Done button to return to the previous installation window . See Figure 6-18.

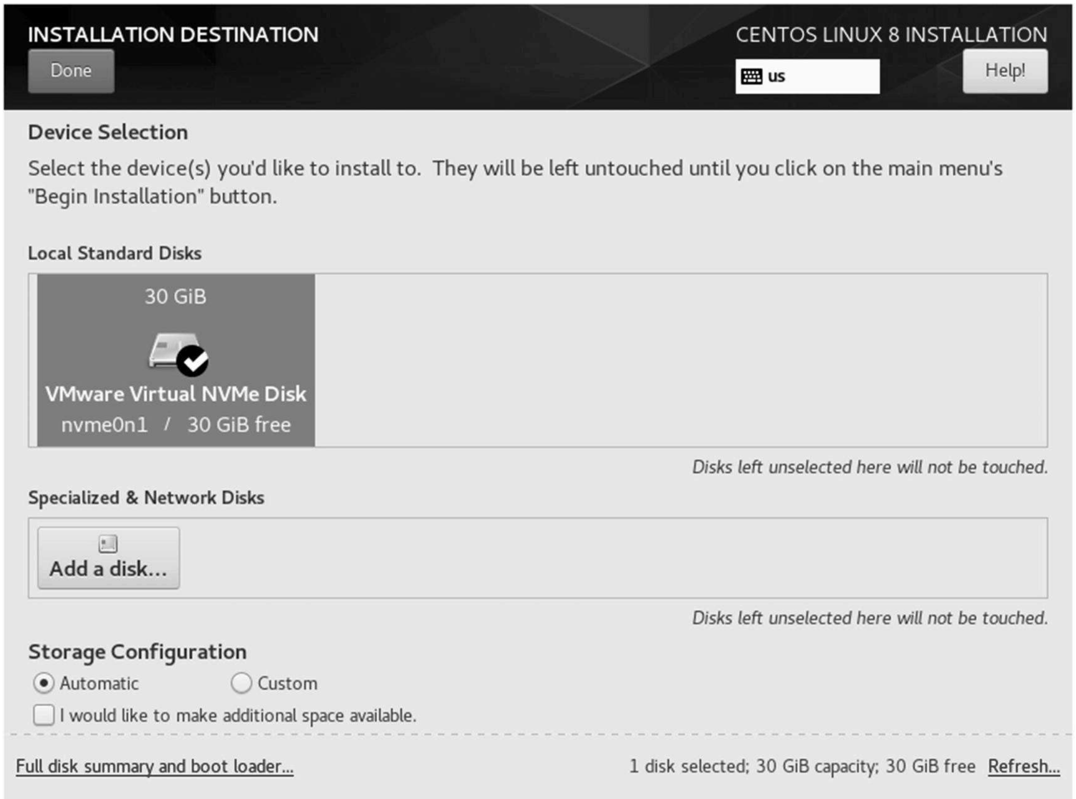


Figure 6-18. CentOS 8 installation, installation destination selection

17 Next, click on “Software Selection Server with GUI” to install extra software during the installation. See Figure **6-19**.

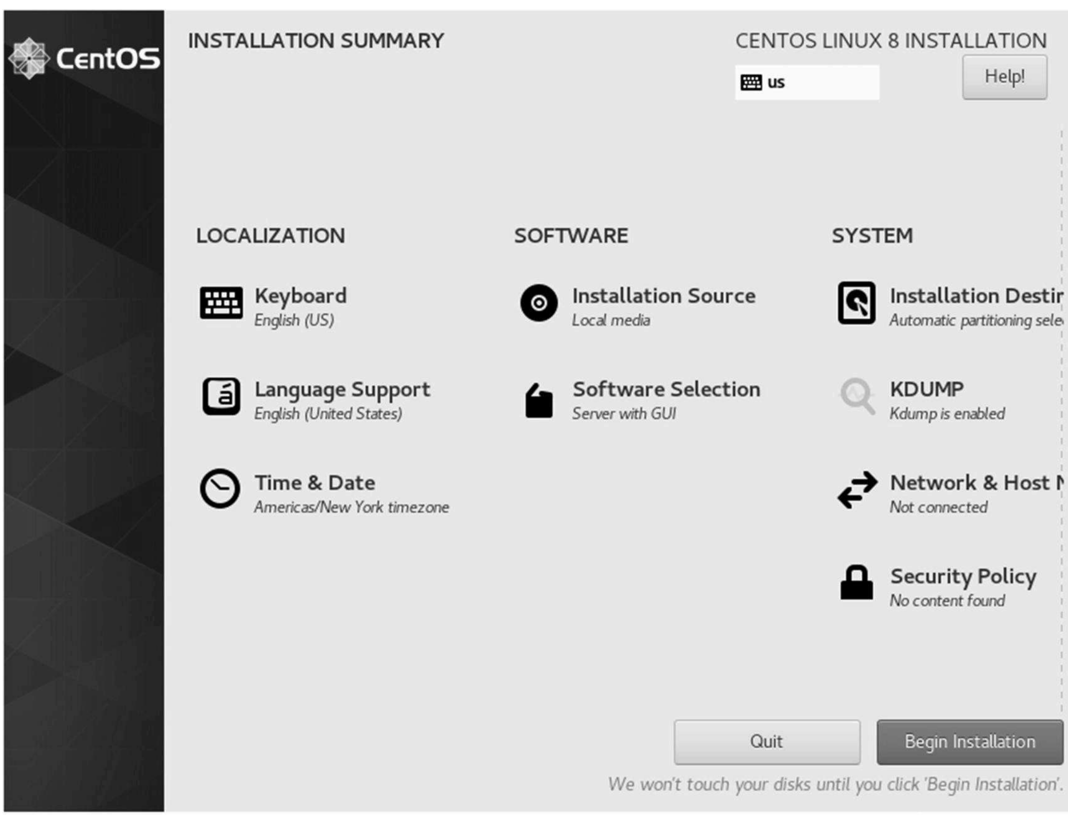


Figure 6-19. CentOS 8 installation, software selection

18 In the Software Selection window, select Server with GUI and select other software such as FTP Server, Mail Server, and more. Here, install at least FTP Server to save time for later. Other IP services will be installed manually in the next chapter. Once you are happy with your selection, click the Done button.

Figure **6-20**, Figure **6-21**, and Figure **6-22** show the additional software you can select under Software Selection.

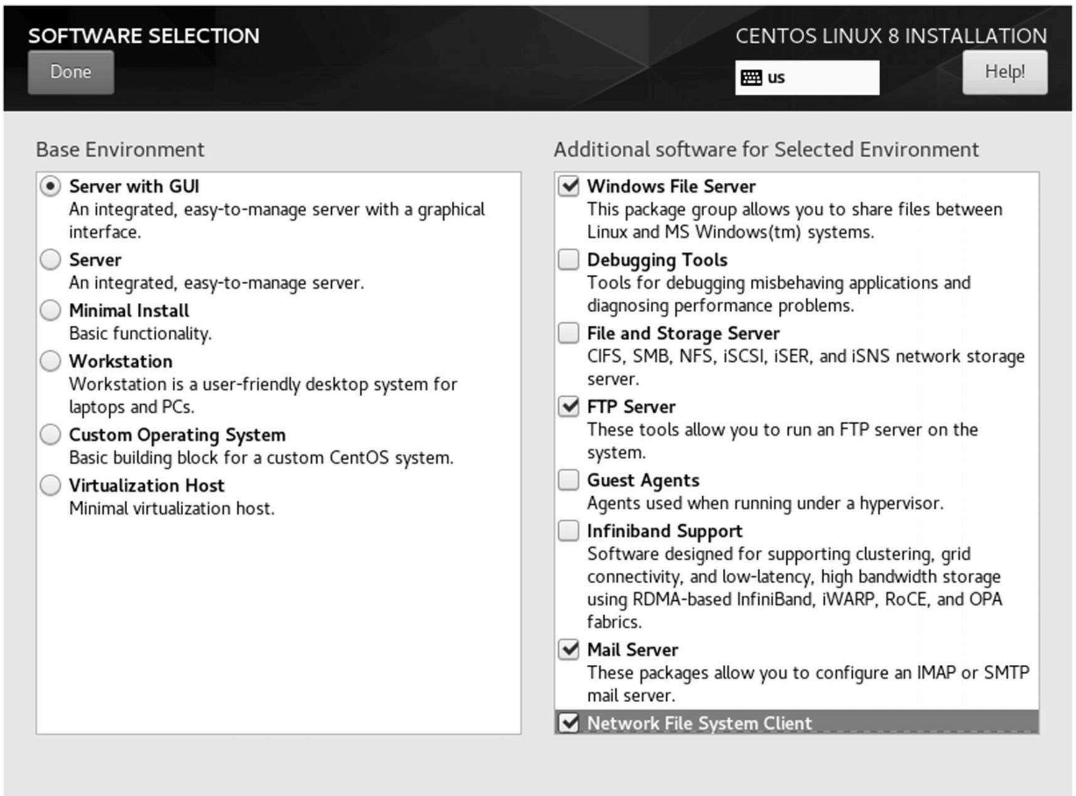


Figure 6-20. CentOS 8 installation, software selection 1

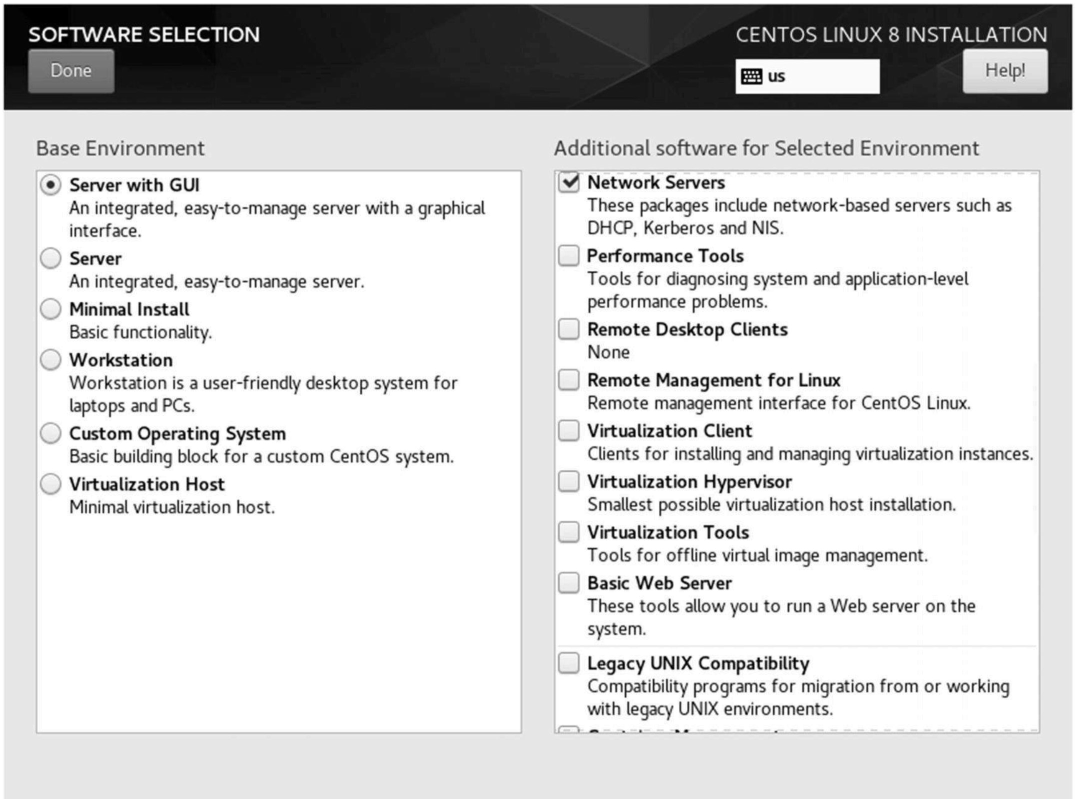


Figure 6-21. CentOS 8 installation, software selection 2

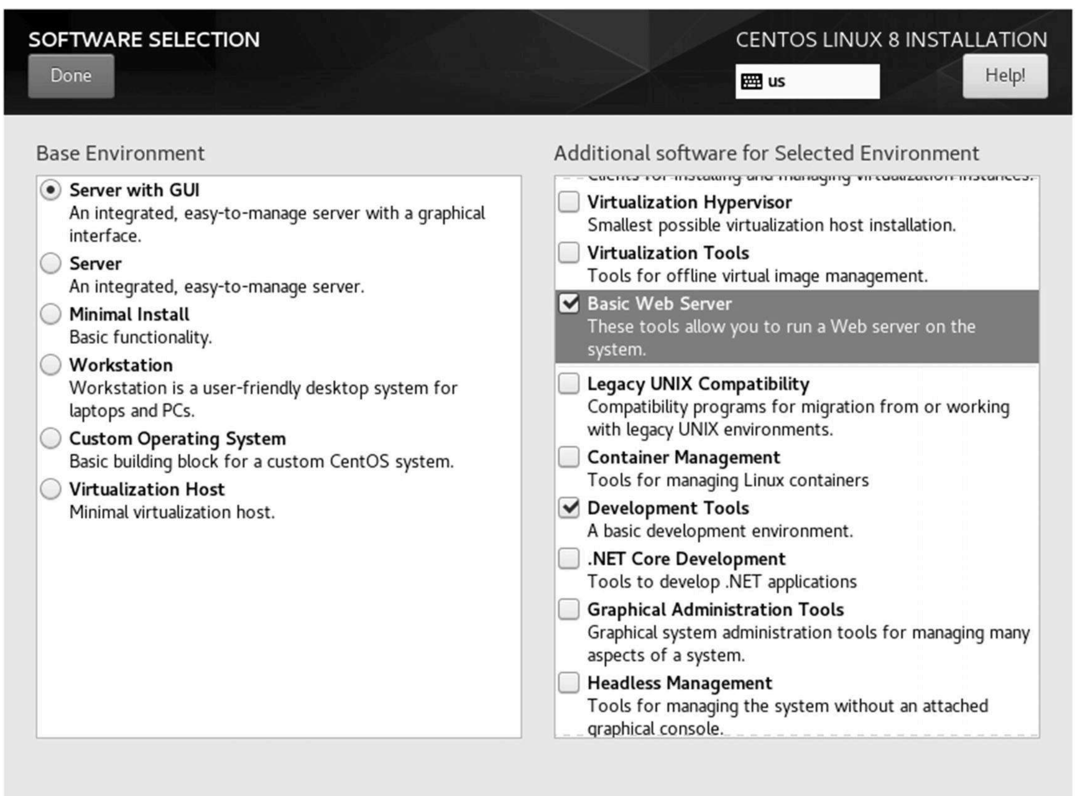


Figure 6-22. CentOS 8 installation, software selection 3

19. Next click Network & Hostname . See Figure 6-23.

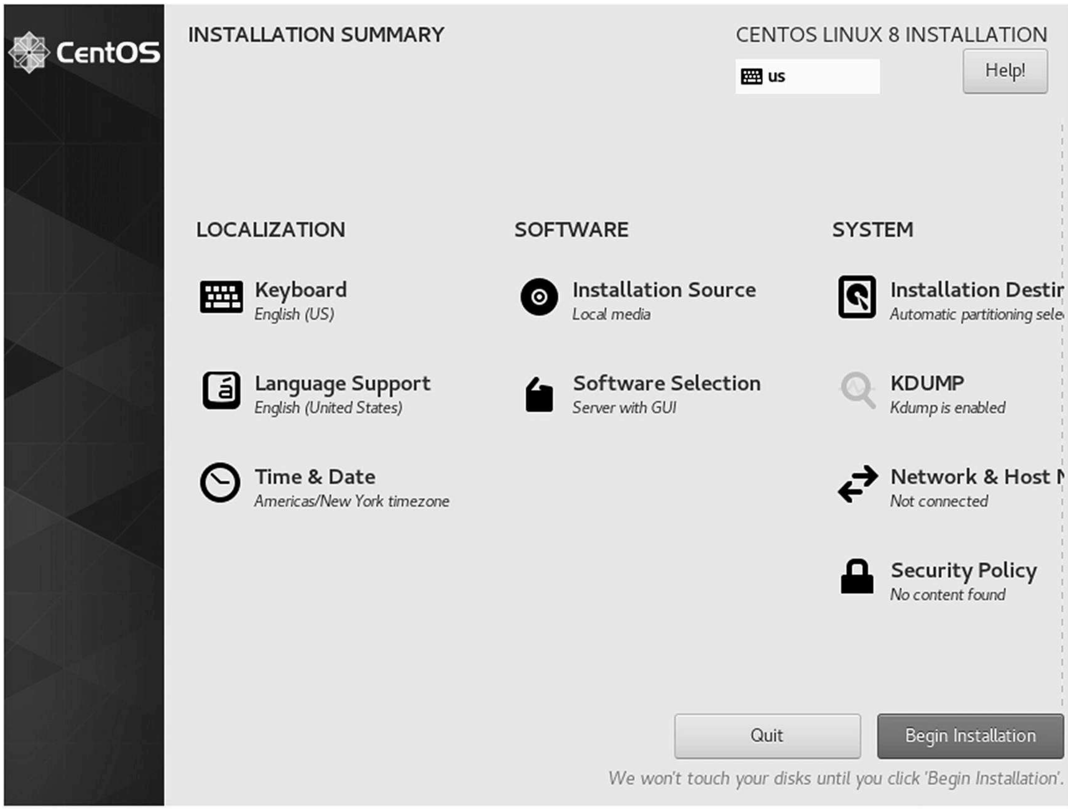


Figure 6-23. CentOS 8 installation, installation summary

20 The network adapter of the CentOS server is turned off by default. Click the ON button to switch on the network adapter . Once the switch is powered on, an IP address will be automatically assigned from the NAT's DHCP subnet. Figure **6-24** received an IP address of 192.168.183.130/24. Once you have confirmed the DHCP allocates the IP assignment, click Done to exit. See Figure **6-24**.

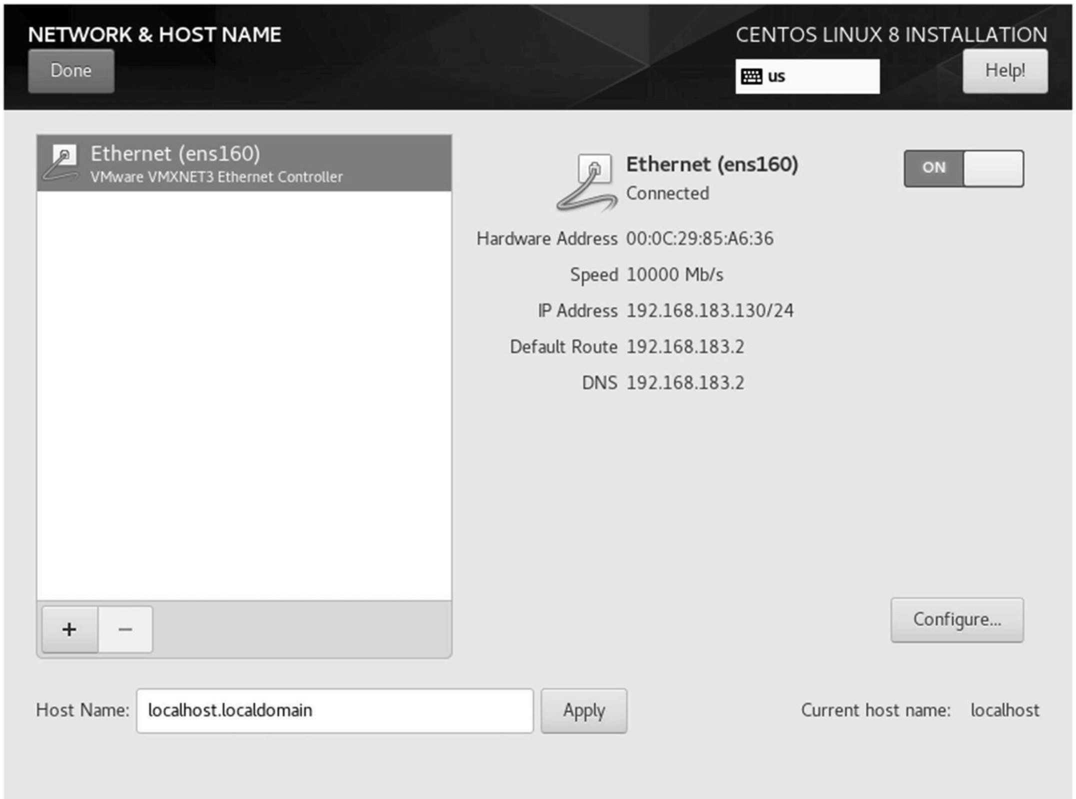


Figure 6-24. CentOS 8 installation, connecting to a network

21 Once you return to the Installation Summary window , click the Begin Installation button to begin the OS installation. See Figure **6-25**.

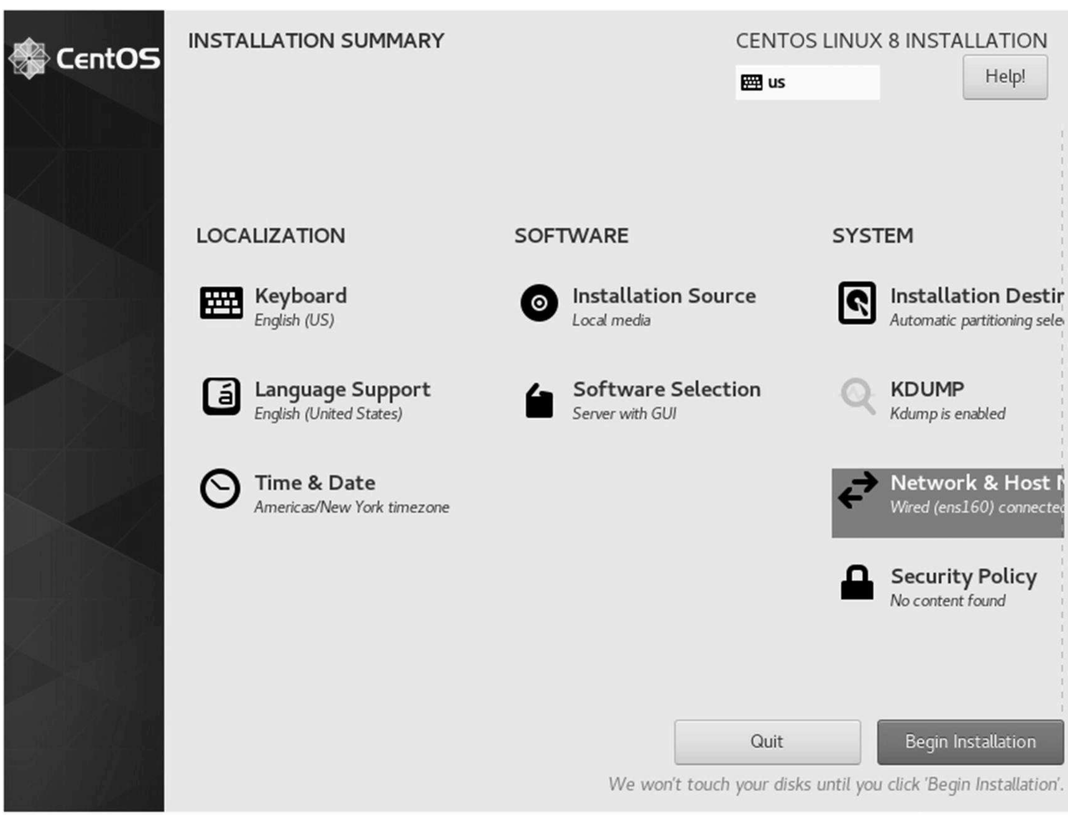


Figure 6-25. CentOS 8 installation, begin installation

22 While downloading packages, you can try to set the root user's password before entering the installation mode. Click Root Password. See Figure 6-26.

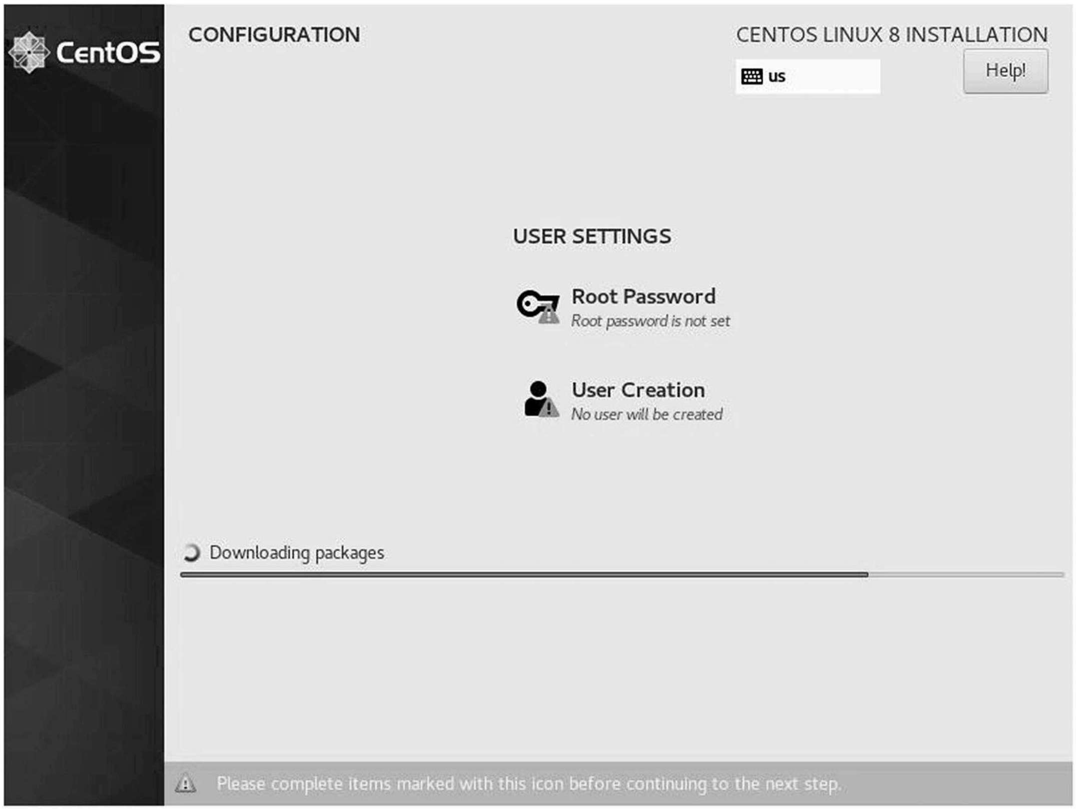


Figure 6-26. CentOS 8 installation, setting up the root user password

23 Enter a root user password twice and click the Done button to exit. See Figure 6-27.

Figure 6-27. CentOS 8 installation, entering the root user password

24 As soon as you return to the previous screen, the CentOS8 server installation will begin. Now, wait for about five minutes for the installation to complete. See Figure **6-28**.

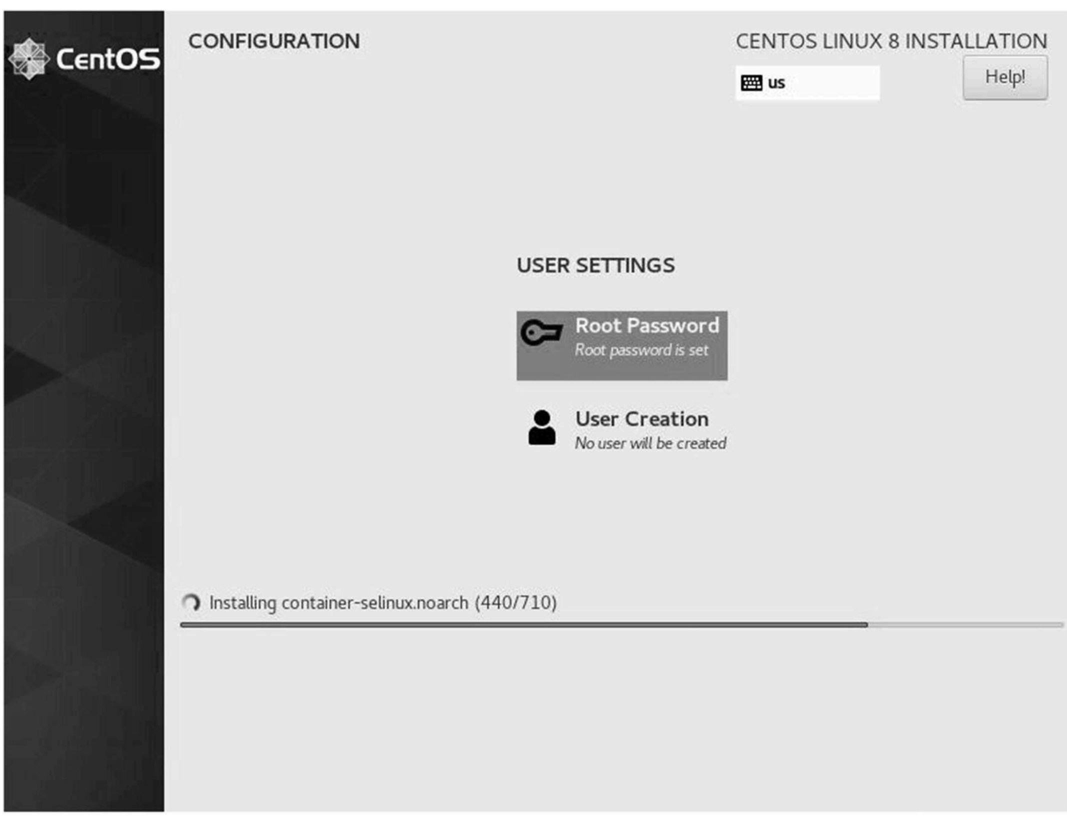


Figure 6-28. CentOS 8 installation, server installation in progress

25 Once the CentOS 8 installation completes, you will be prompted for a reboot. Click Reboot to restart your new Linux server. See Figure **6-29**.

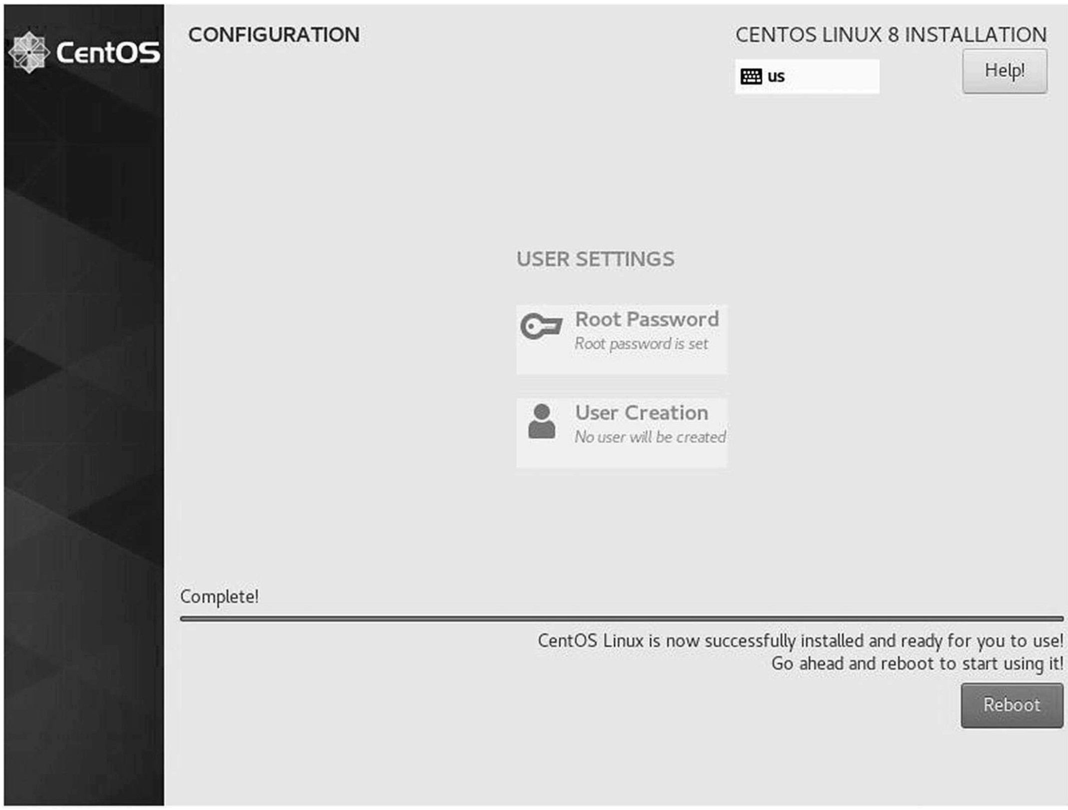


Figure 6-29. CentOS 8 installation, installation completed and reboot

26 After rebooting the server, you will see an initial configuration screen . Click License Information to agree to the end-user license agreement to use CentOS for free. See Figure **6-30**.

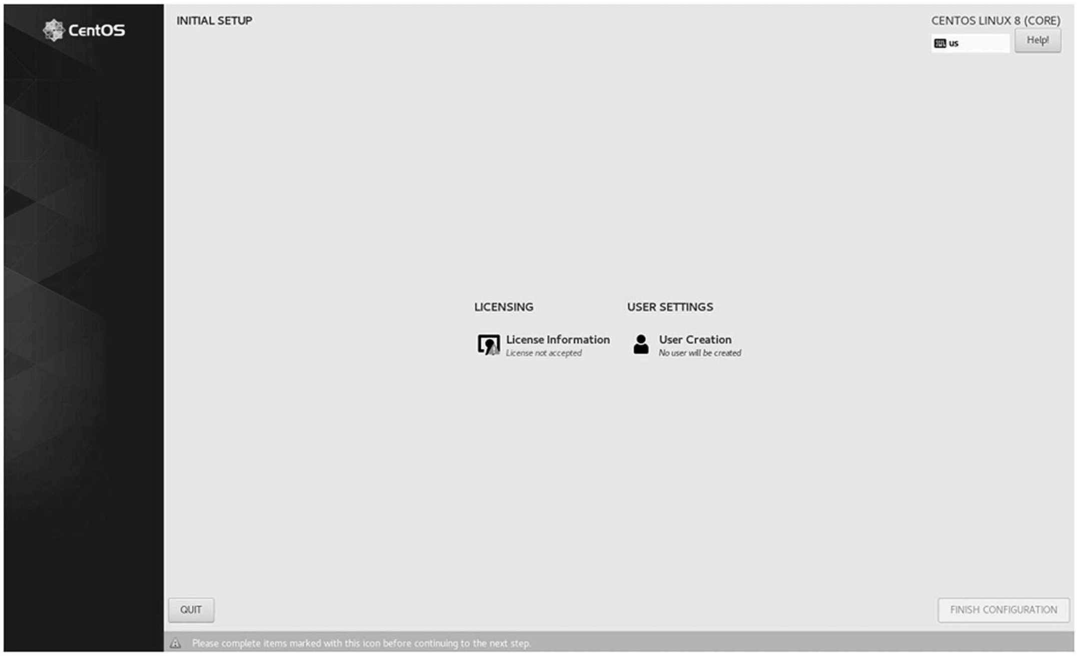


Figure 6-30. CentOS 8 installation, initial configuration window

27 Click the "I accept the license agreement " checkbox and click Done. See Figure **6-31**.

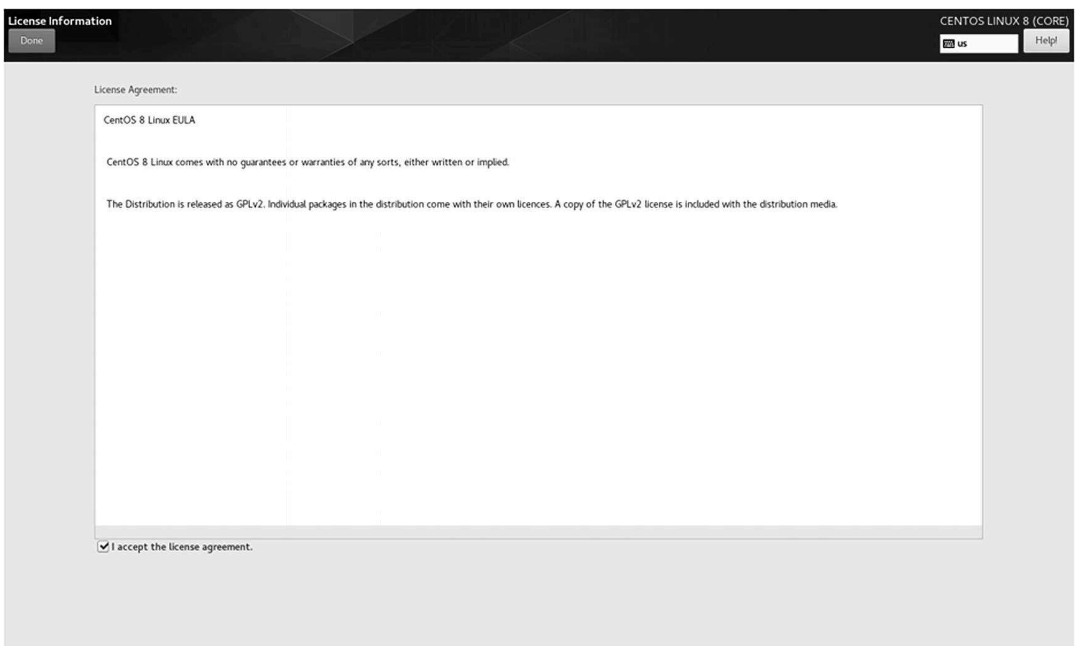


Figure 6-31. CentOS 8 installation, accepting the end-user license agreement

28 When you return to the initial setup window , you notice that the warning sign has disappeared in the License Information field. Now click the Finish Configuration button to log into CentOS for the first time. See Figure **6-32**.

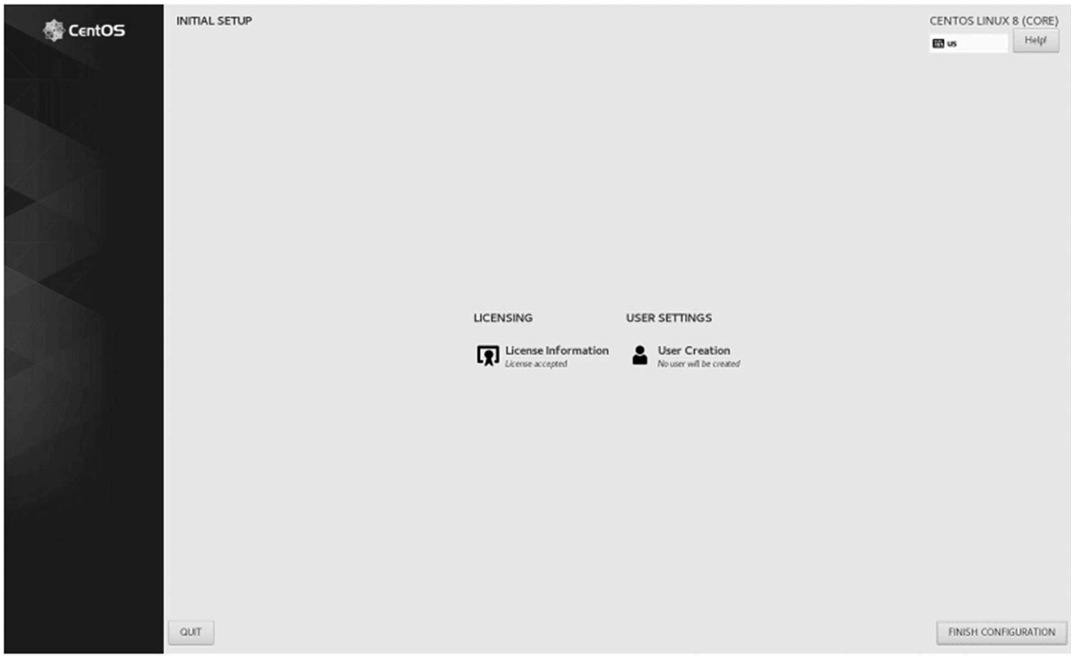


Figure 6-32. CentOS 8 installation, initial setup completed

29 Click Next. See Figure **6-33**.

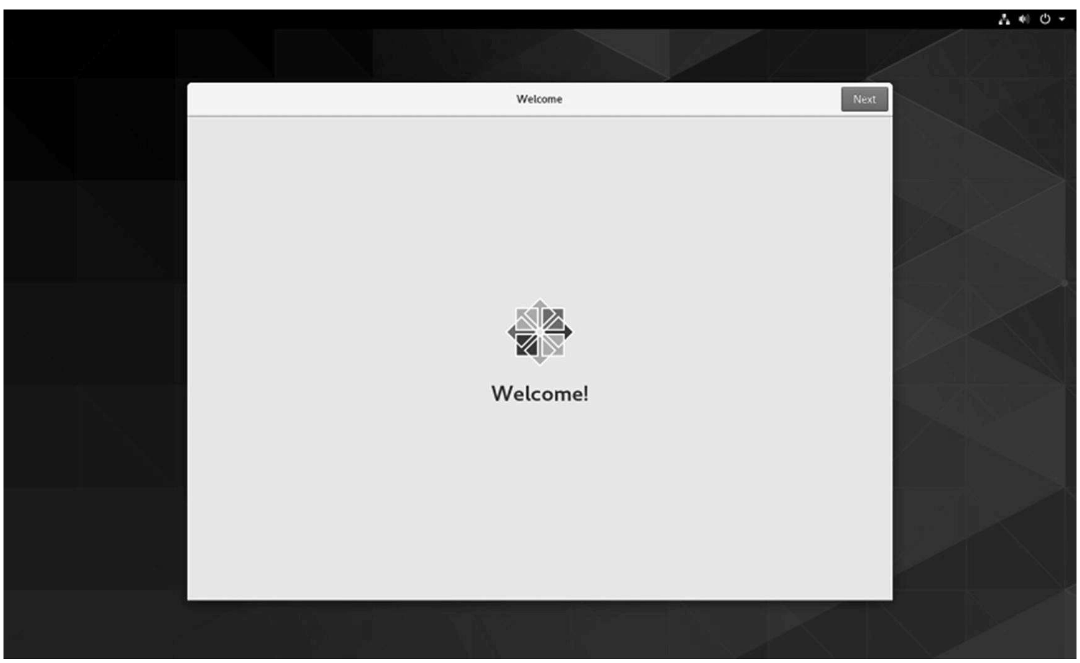


Figure 6-33. CentOS 8 installation, welcome window

30 Click the Next button. See Figure **6-34**.

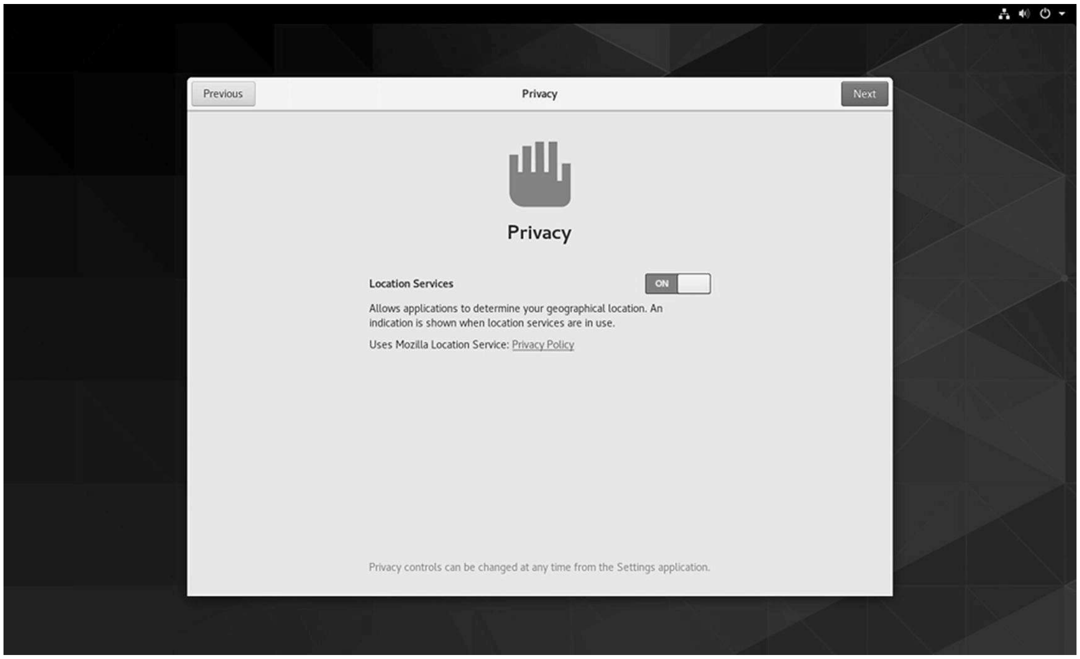


Figure 6-34. CentOS 8 installation, privacy location services

31 Click the Skip button to skip the online accounts configuration . See Figure **6-35**.

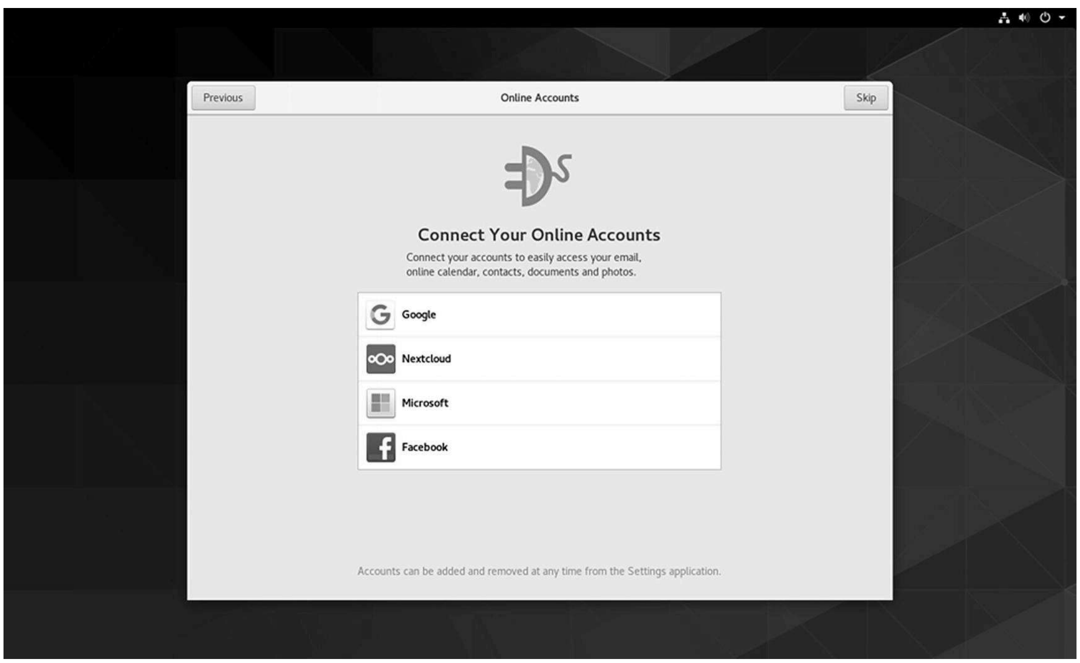


Figure 6-35. CentOS 8 installation, online accounts

32 Enter your details and click Next . See Figure **6-36**.

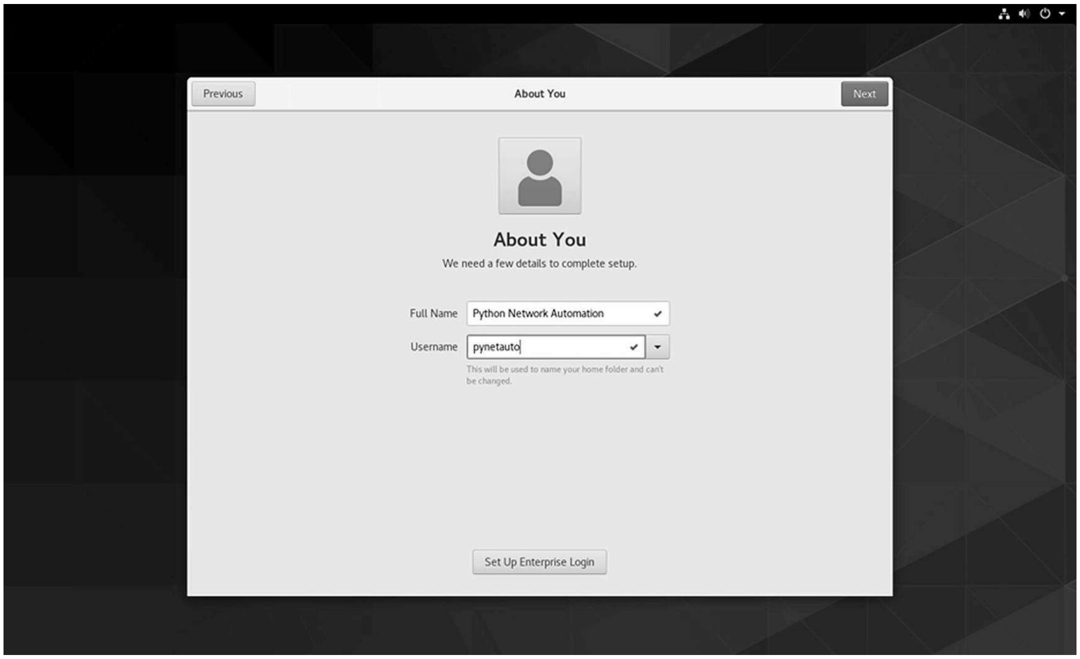


Figure 6-36. CentOS 8 installation, About You screen

33 You will be taken to the Password screen next . Enter your password twice and click the Next button. See Figure **6-37**.

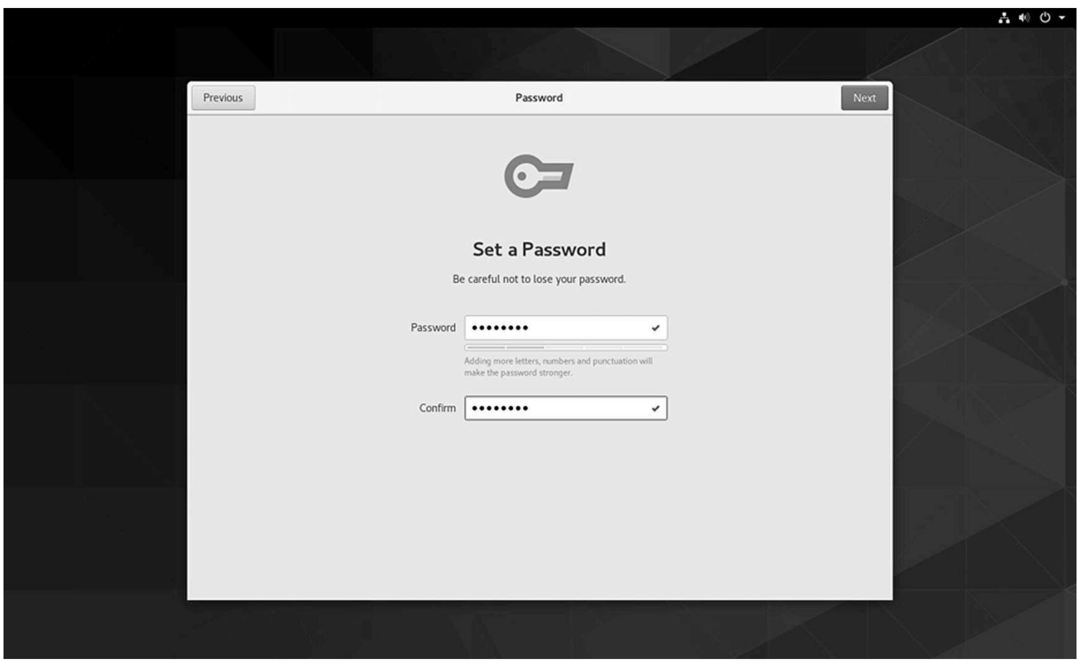


Figure 6-37. CentOS 8 installation, setting a password

34 On the Ready to Go screen , click Start Using CentOS Linux to complete the configuration. See Figure **6-38**.

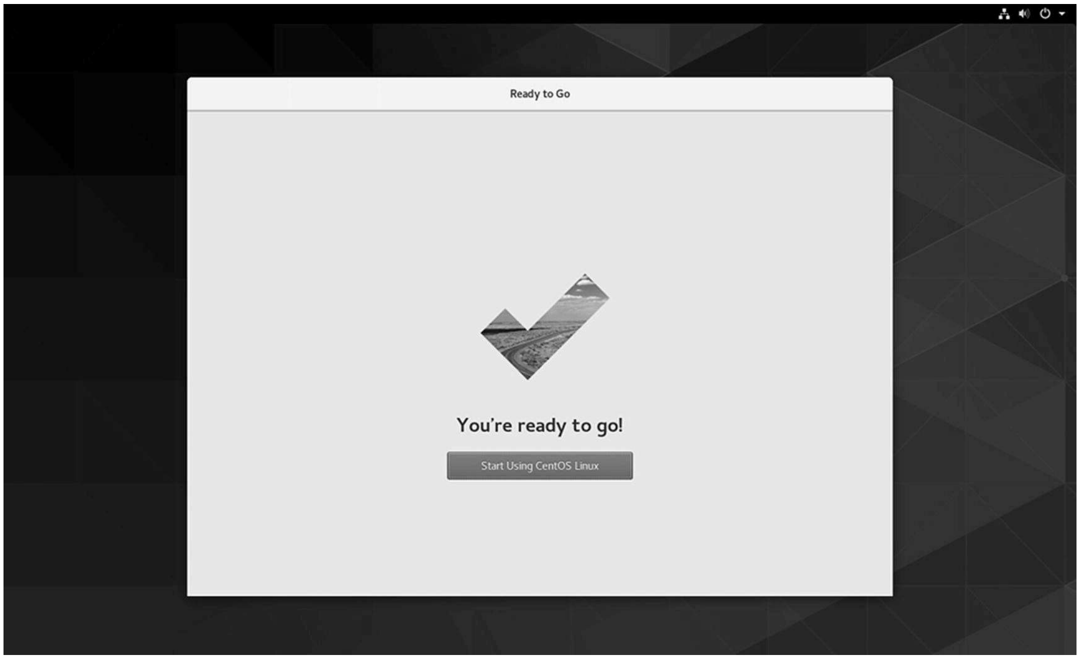


Figure 6-38. CentOS 8 installation, Ready to Go screen

35 Enter your password and click the Sign In button . See Figure **6-39**.

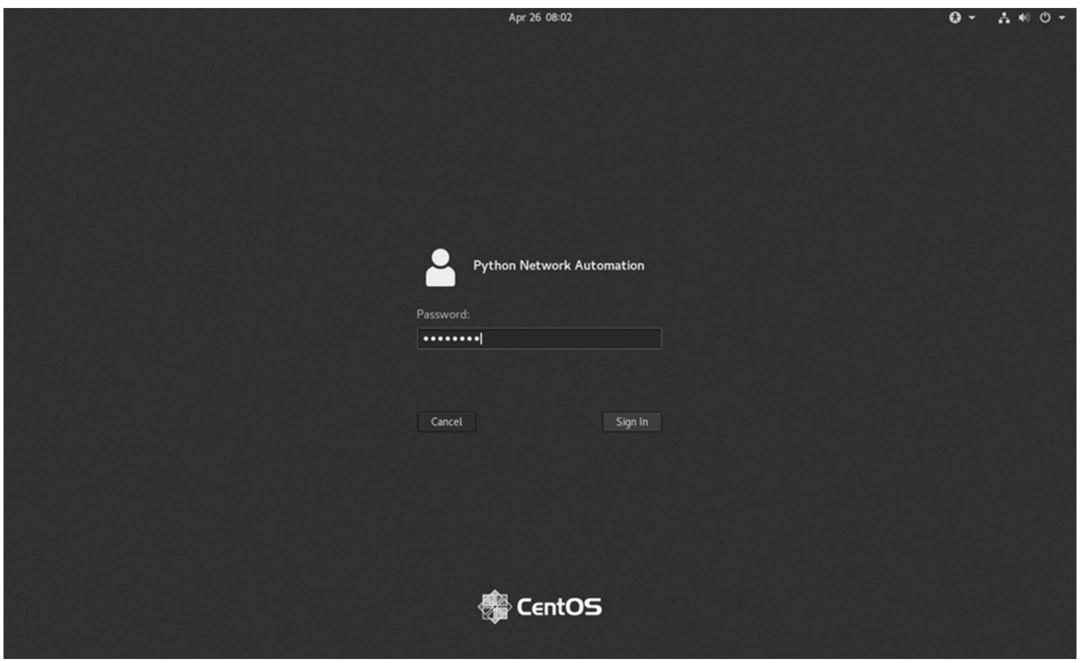


Figure 6-39. CentOS 8 installation, first sign-in

36 You will now see the Getting Started screen . It is safe to close the window and begin using your CentOS server now. See Figure **6-40**.

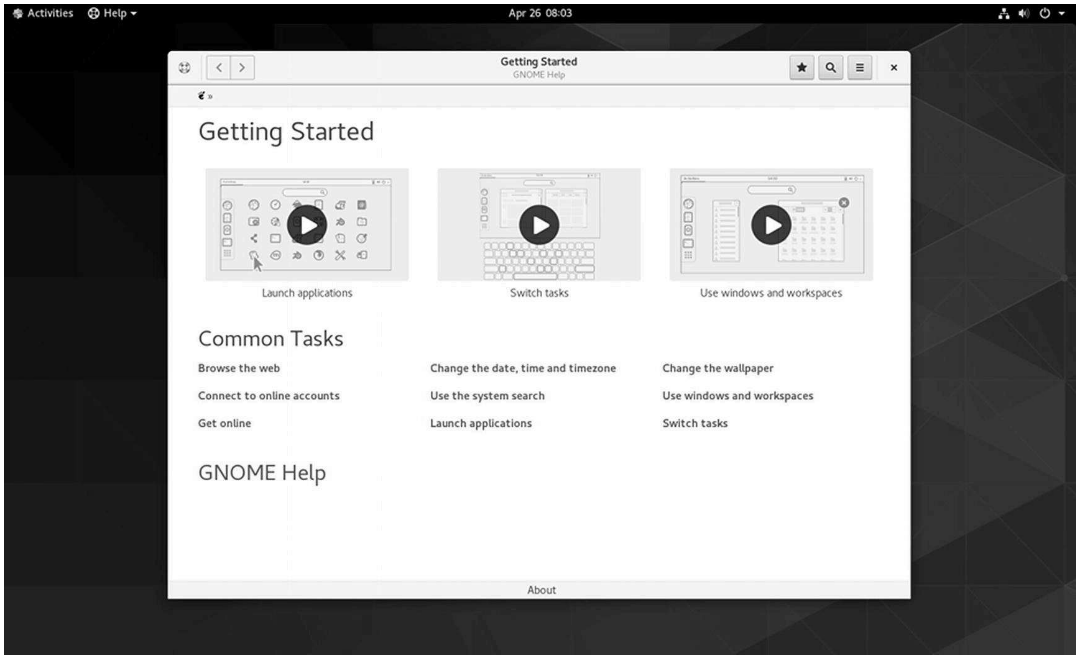


Figure 6-40. CentOS 8 installation, password configuration

37 Launch an application such as the Terminal Console to use the server. See Figure **6-41**.

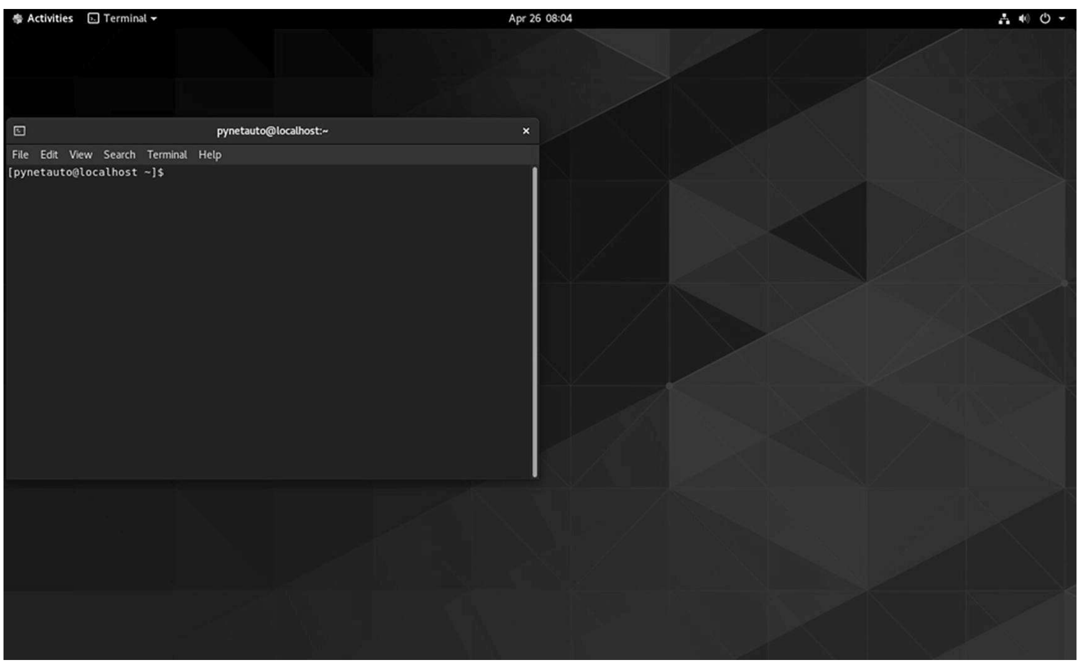


Figure 6-41. CentOS 8 installation, first-time use

Logging In to a New CentOS 8 Server via SSH

On CentOS 8, you do not have to change any settings to connect to the server remotely using an SSH connection; both the user and root user will be allowed to connect to the server using the SSH protocol by default. Do not take my word for this. Let's quickly validate this by opening up PuTTY on your host Windows 10 and connecting via SSH into the host Windows 10 desktop server.

#	Task

1 First, from your host's Windows desktop, open PuTTY and enter the IP address of your CentOS 8 VM. The example shown here uses an IP address of 192.168.183.130, but your IP address may be different, and that is still OK. Click the Open button. See Figure **6-42**.

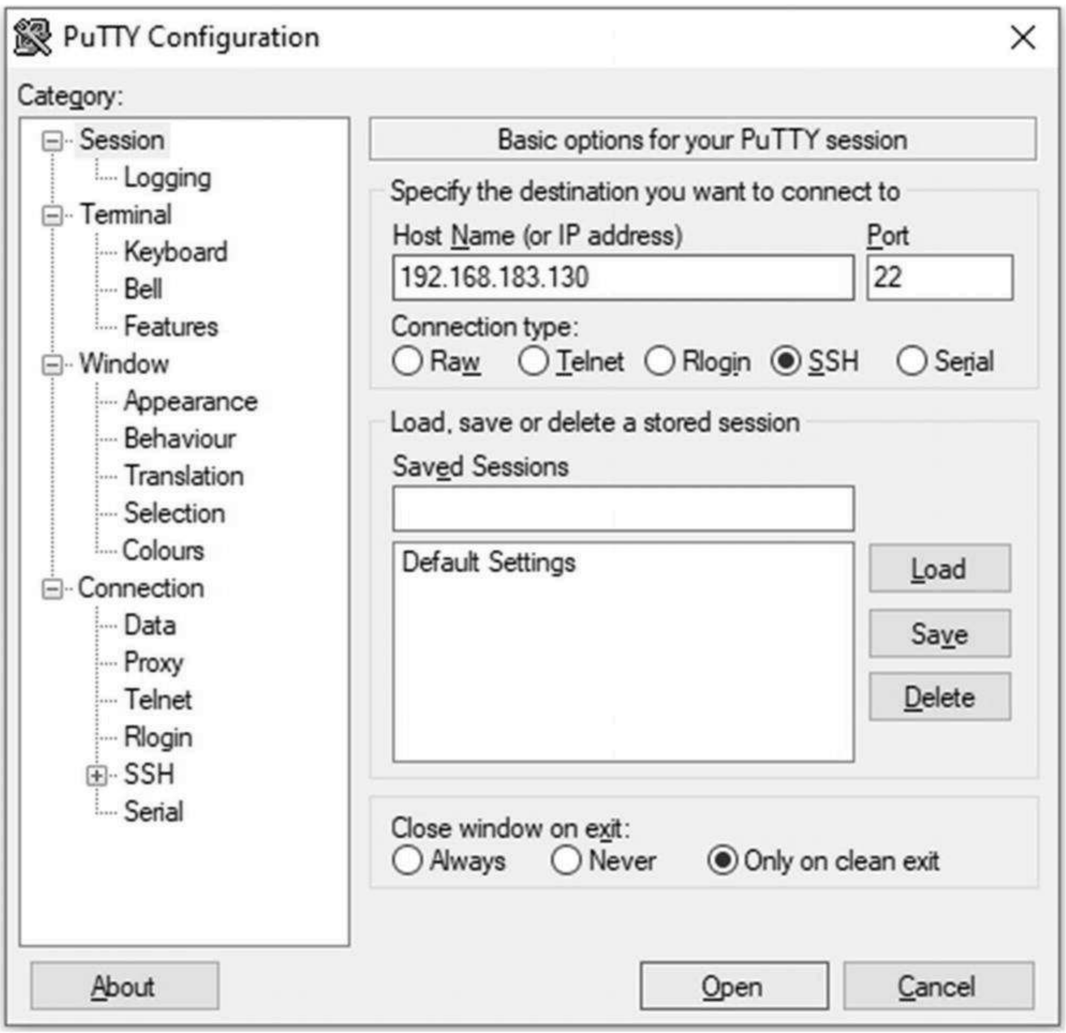


Figure 6-42. CentOS 8 SSH login, PuTTY SSH login to CentOS 8

2 When the PuTTY security alert is prompted, accept the rsa2 key , and click the Yes button. See Figure **6-43**.

Figure 6-43. CentOS 8 SSH login, accepting the rsa2 server key

3 Now use your user ID and password to confirm that the SSH login works . See Figure **6-44**.

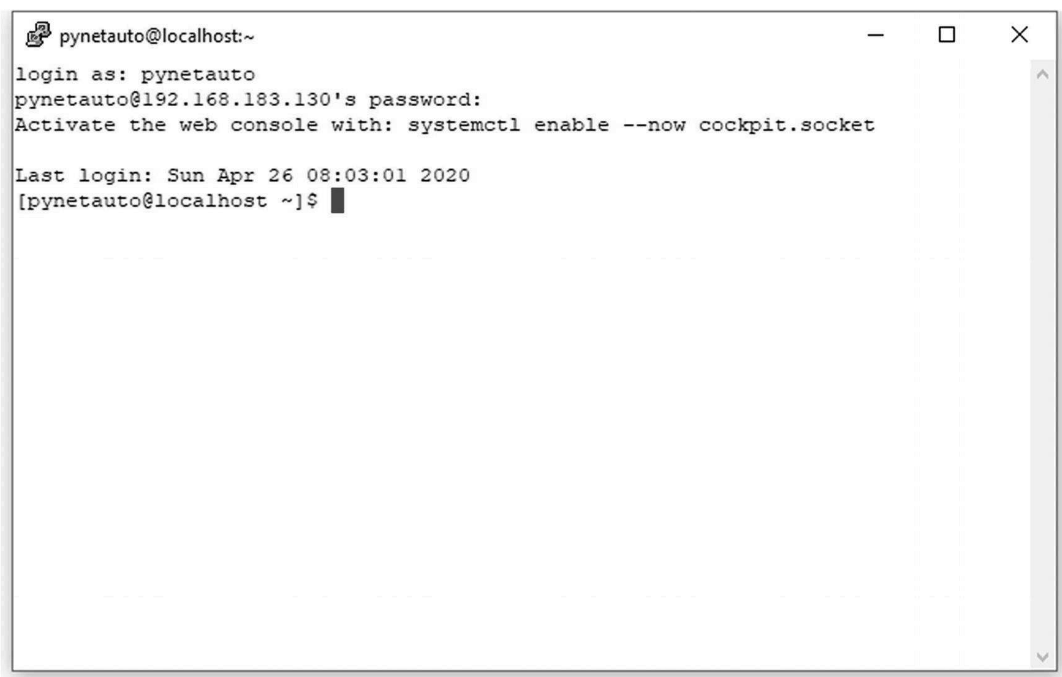


Figure 6-44. CentOS 8 SSH login, user login

4 Optionally, start another PuTTY session and, this time, log in as the root user. You have made no system changes, but CentOS 8 will allow you to connect to the server using the SSH protocol. As you will discover, just like

Ubuntu , CentOS has real potential to become one of your favorite Linux distro systems. See Figure **6-45**.

Figure 6-45. CentOS 8 SSH login, root user login

This completes the CentOS 8 installation and initial setup. Before moving onto the next topic, here are some of my thoughts on the Linux operating system.

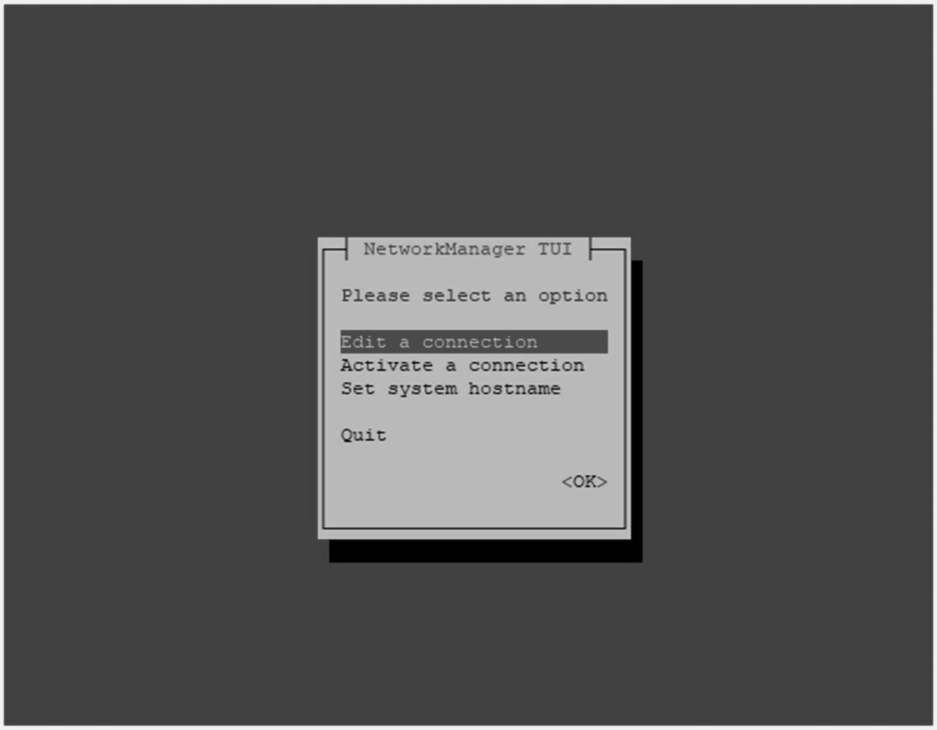
Quick Internet research reveals that the total percentage of PC users preferring the Linux operating system over Windows OS is only a minor percentage; however, at the enterprise level of IT services, the percentage of Linux OS used is almost equal to Microsoft Windows OS, and the popularity of the Linux system is gaining momentum. If you are a new IT technician entering the IT industry, this is actually important news as more opportunities are offered to those who can tame and master Linux as their preferred OS at the enterprise level.

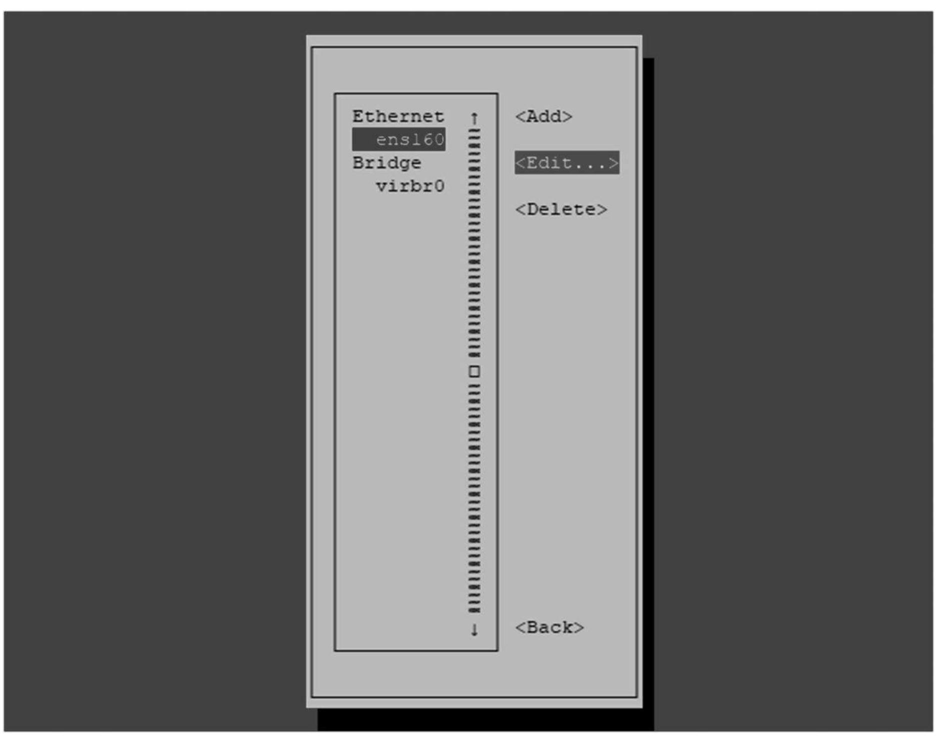
I have been a network engineer for many years now. I admit I have gotten too cozy with my Windows desktop and server operating systems; I have also been guilty of refusing to open my mind and start learning Linux. This changed as soon as I started teaching myself Python as I was getting frustrated and struggling to do more using Windows OS . After studying Python, I think network automation using Python is easier on Linux OS than Windows OS; hence, if you are serious about making Python your profession, start learning Linux. For network automation using Python, Linux knowledge is a prerequisite, not an option. Learn Python and Linux at the same time based on your substantial networking knowledge and experience.

Managing a Network Adapter on Linux

If you want to change the network settings on your Linux machine, for example, give your CentOS server a static IP address, so the IP address does not change after the DHCP lease expires. You can easily change this setting using the `nmtui` tool. If you forgot to power on the network adapter during Linux installation, you could change the settings using this tool. Quickly run the `sudo nmtui` command from your CentOS. If you are logged in as a `sudo` user, you can omit `sudo` in front of your command.

#	Task
01	You can run the `nmtui` command from the VMware Workstation Console or PuTTY SSH session. Type in `sudo nmtui` and hit the Enter key on your keyboard. See Figure **6-46**.  ***Figure 6-46.*** CentOS 8 configuration, sudo nmtui command
02	When CentOS's NetworkManager TUI appears, select "Edit a connection." See Figure **6-47**.

#	Task
	 *Figure 6-47.* CentOS 8 configuration, NetworkManager TUI
03	Select ens160 and click Edit . See Figure **6-48**.



#	Task

Figure 6-48. CentOS 8 configuration, selecting a network adapter

04 Click the Automatic option on the right side of IPv4 Configuration and change it to Manual. Click the Show option on the right. See Figure **6-49**.

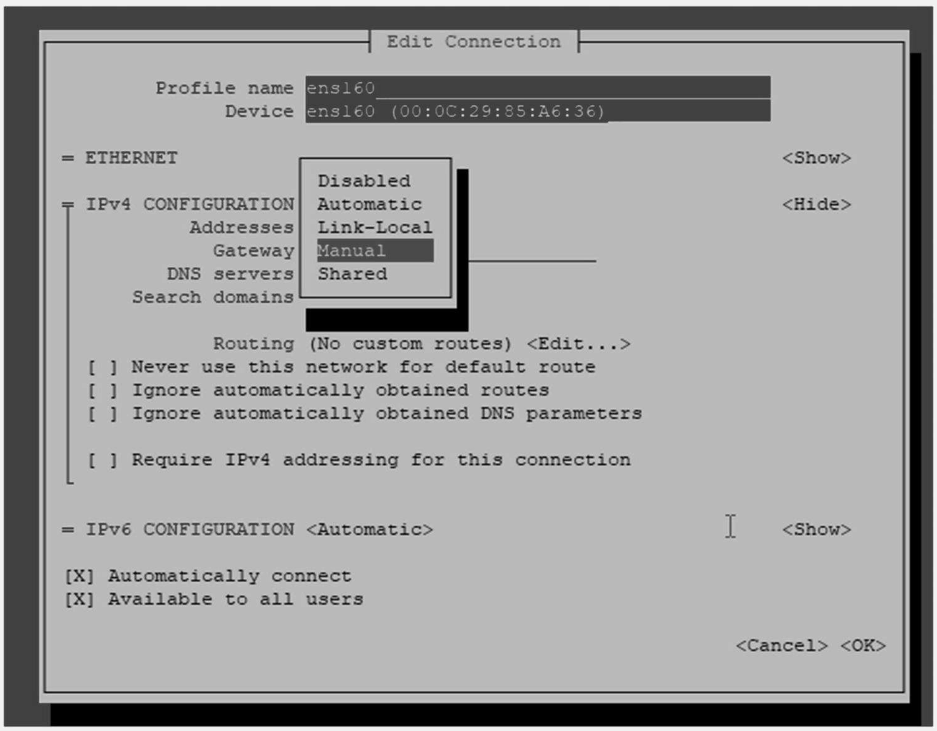


Figure 6-49. CentOS 8 configuration, selecting a manual configuration

05 Now enter your server IP address and the NAT gateway's IP address . Move down and click OK. See Figure **6-50**.

#	Task

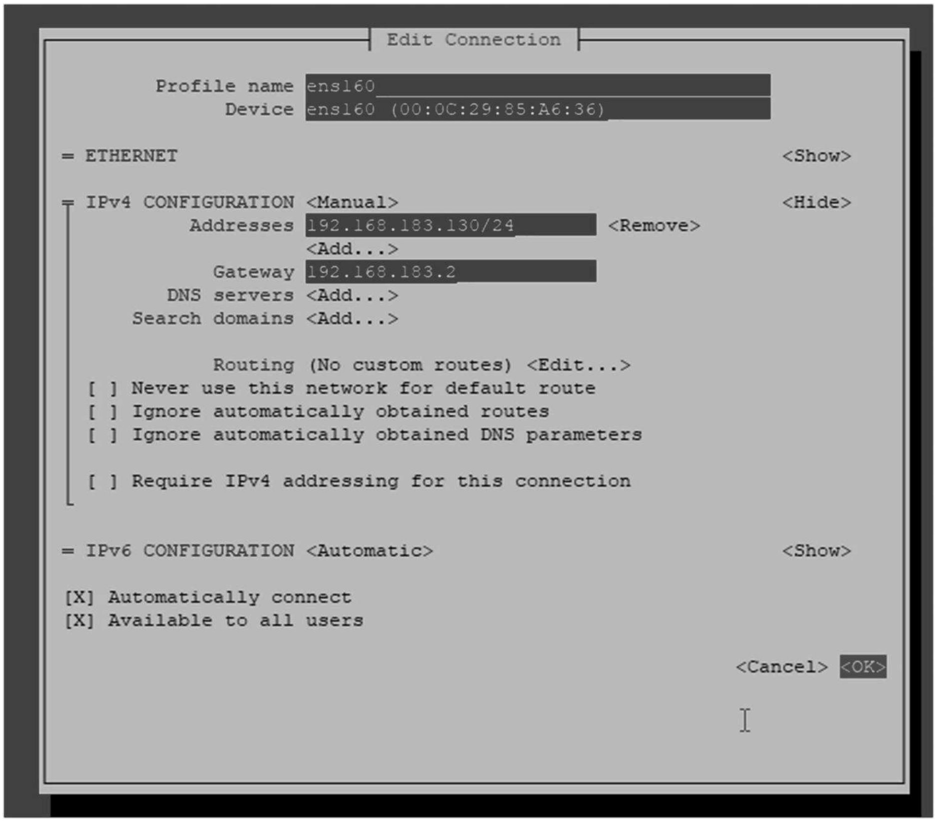
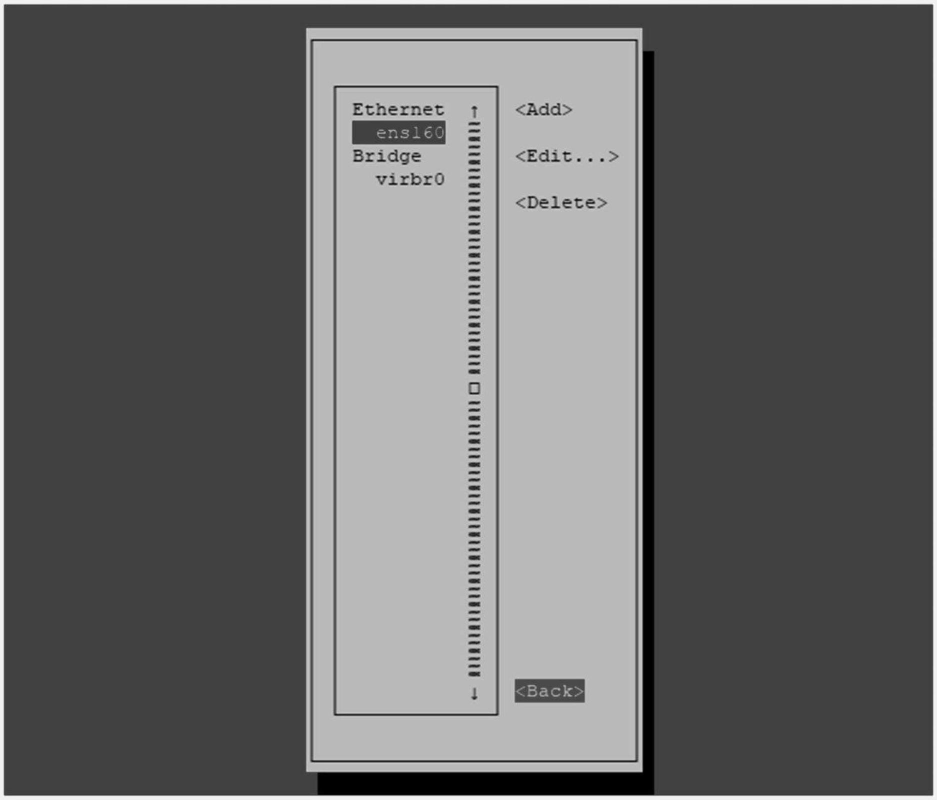


Figure 6-50. CentOS 8 configuration, saving manual changes

06	Press the Tab key to move down and select Back to return to the first TUI interface. See Figure **6-51**.

#	Task
	 ***Figure 6-51.*** CentOS 8 configuration, Back option
06	Click Quit to save your changes and exit the application . See Figure **6-52**.

#	Task

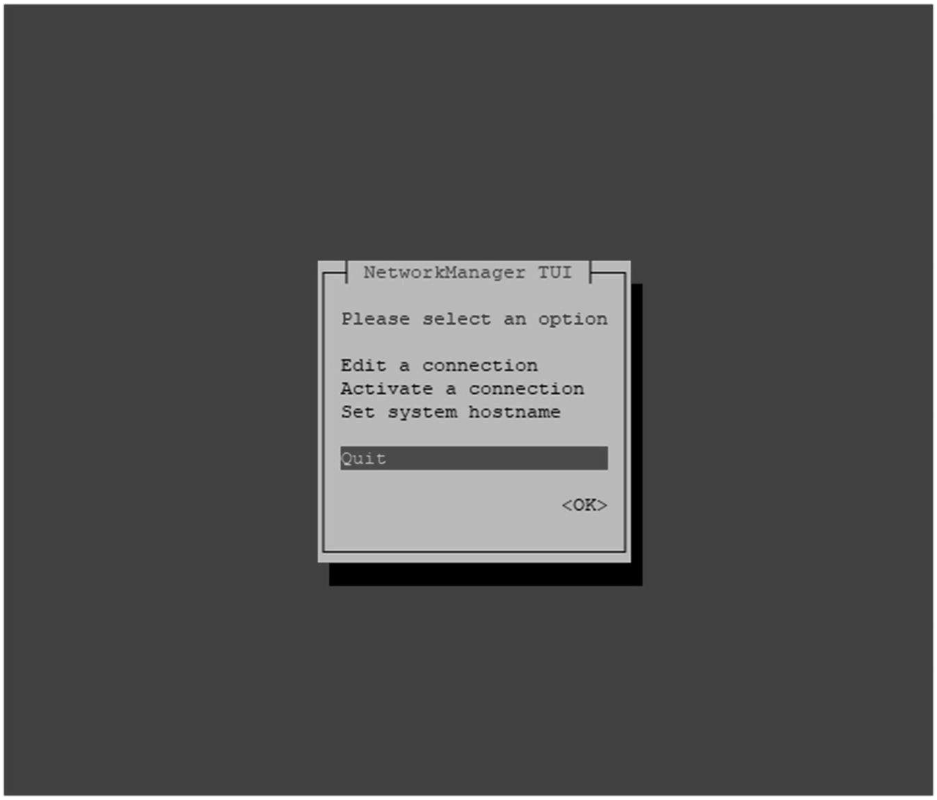


Figure 6-52. CentOS 8 configuration, saving and exiting TUI

At this stage, consider how you are going to use this server in your lab and install any software. Once you are satisfied with the current state of your CentOS 8.1, Linux VM customization takes a snapshot of the VM while running or makes a full clone of powered-off VM files. Taking a snapshot or cloning is the same as in the previous Ubuntu 20 process. Later in this chapter, you will create a third VM from an `ova` file, which is another way to create your VMs. An `ova` file is a virtual appliance used by VMware Workstation and Oracle VM VirtualBox. An `ova` file is a packaged file that contains files used to describe a VM, which includes an `ova` descriptor file, optional manifest (.MF) and certificate files, and other related files. In other words, an `ova` file is a predefined VM template that another VM administrator has prepared in advance to deploy a specific server instantly. You will download a GNS3 VM `ova` file from the GNS3 GitHub site and create a GNS3 VM by importing the file; this is a preparation step for GNS3 installation and integration, which will take place in Chapter **10**. So, do not skip the GNS3 VM creation process here.

Creating a GNS3 VM by Importing an .ova File

A GNS3 VM `.ova` file is maintained by community developers at **www.gns3.org** and is used to create a VM that hosts network, security, and system images running on the GNS3 application. This server is a Linux server purposefully built to offload the workload of the physical host of GNS3, so the emulated GNS3 devices run smoothly without affecting the underlying physical host. You will soon find out that the GNS3 VM offers a lot more than running routers and switches. It can also host containerized Linux servers and more. GNS3 and the GNS3 VM are open source and free to everyone. In this book, we are using VMware Workstation on Windows 10, so we need to download the GNS3 VM files that work on Windows. When writing this chapter, the latest GNS3 version was v2.2.8, and this version will be used here, but try to download the latest version of GNS3 files as the newer versions always offer performance improvements, and many of the bugs have been ironed out. Here we will use GNS3 v2.2.8 so that we will download the corresponding `GNS3.VM.VMware.Workstation.2.2.8.zip` file. You will be reminded to download the `GNS3-2.2.8-all-in-one.exe` file again, but you can also download it now to save time.

Downloading and Installing the GNS3 VM from an .ova File

The process for downloading the GNS3 VM file and creating a GNS3 VM is as follows:

#	Task
01	First, go to GNS3's GitHub site specified here to download the GNS3 VM file. Download the latest version of the GNS3 VM file . The latest version is `GNS3.VM.VMware.Workstation.2.2.8.zip` at the time of

#	Task
	writing this book. At the time of writing this book, the latest GNS VM release version is 2.2.8. See Figure **6-53**.

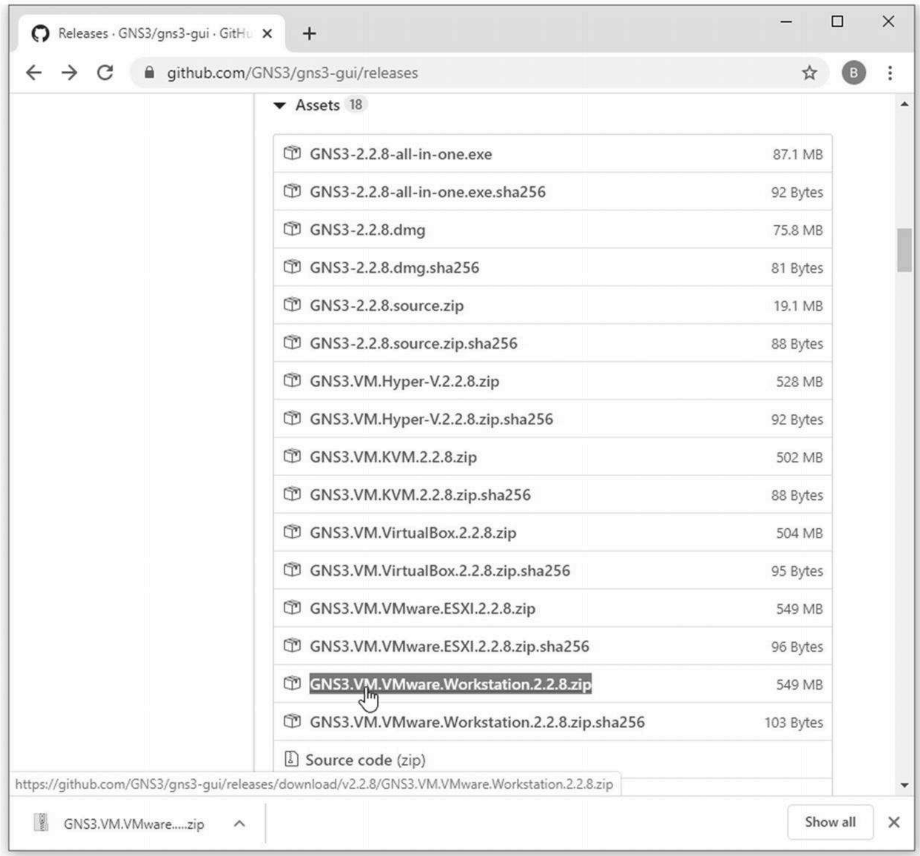
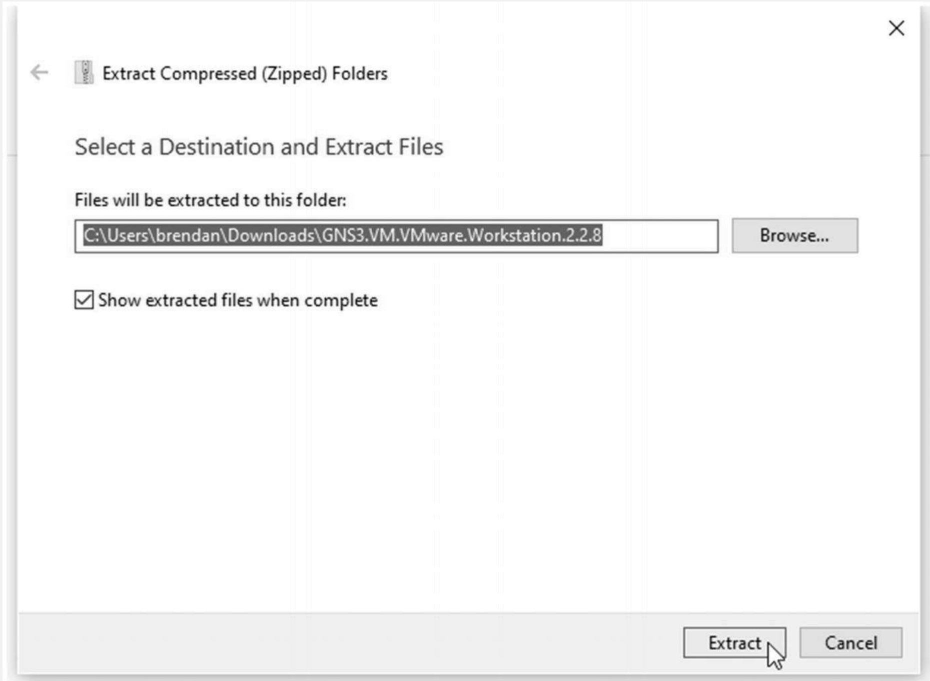
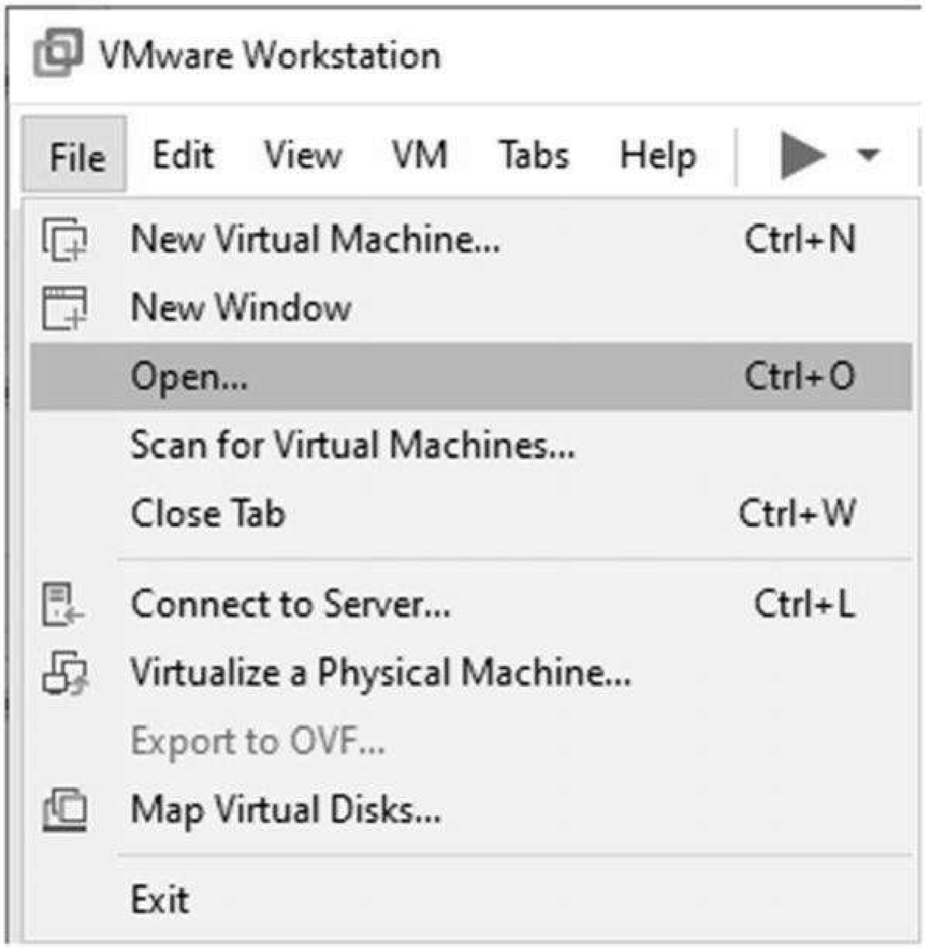


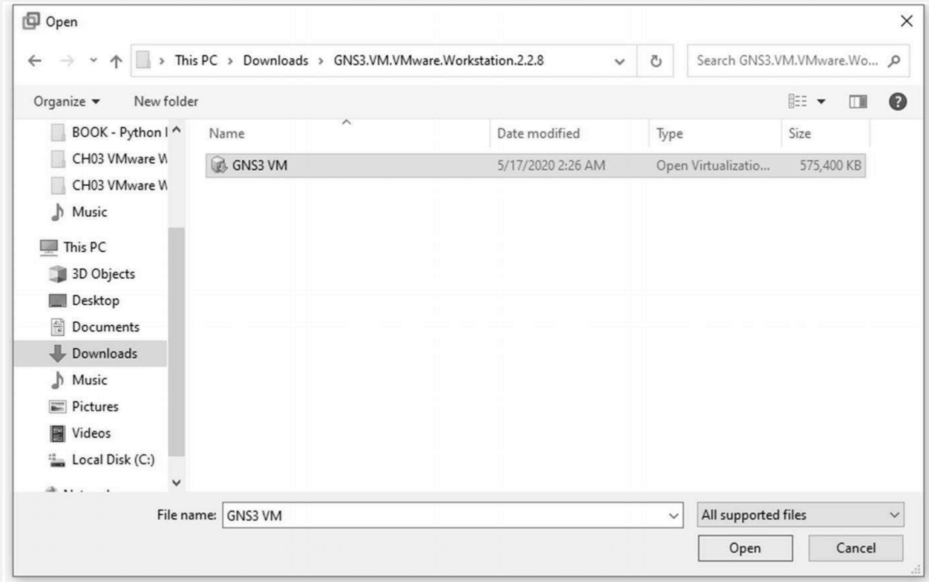
Figure 6-53. GNS3 and GNS3 VM, downloading the .ova file from GitHub

`https://github.com/GNS3/gns3-gui/releases`

02	Now go to your `Downloads` folder and unzip the downloaded file in the same location. See Figure **6-54**.

#	Task
	 ***Figure 6-54.*** Unzipping the GNS3 VM .ova file to a folder
03	Now go back to your VMware Workstation 15 Pro and select the File ➤ Open menu. See Figure **6-55**.

#	Task
	 ***Figure 6-55.*** VMware Workstation menu, File ➤ Open
04	Locate the extracted GNS3 VM `.ova` file , select the file, and then click Open. See Figure **6-56**.

#	Task
	 *Figure 6-56.* VMware Workstation, GNS3 VM .ova file open
05	In the next window, click the Import button to import the GNS3 VM. You can leave the storage path as the default location. See Figure **6-57** and Figure **6-58**.

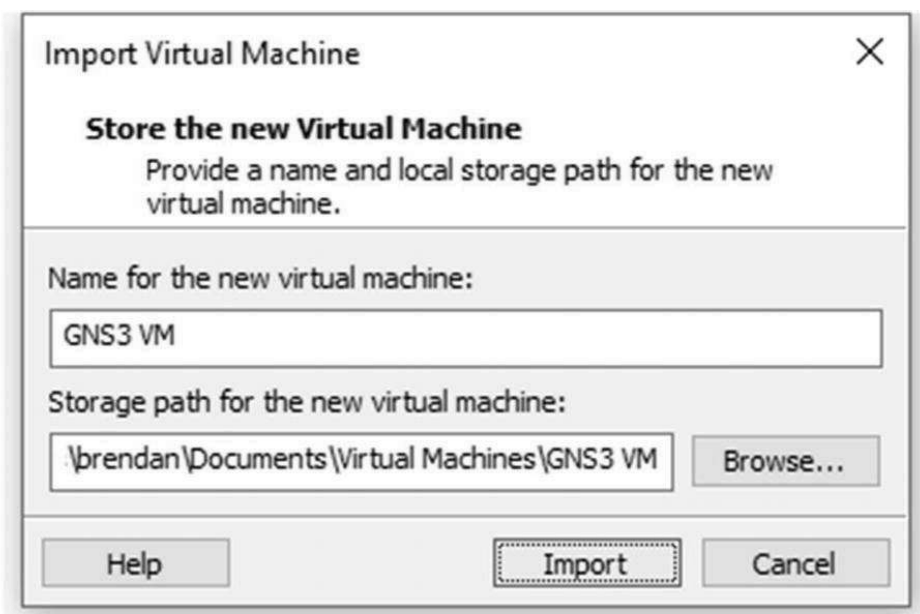


Figure 6-57. VMware Workstation, GNS3 VM import

#	Task

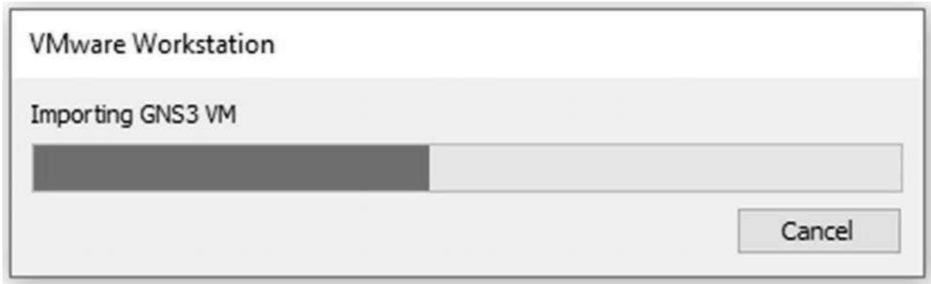


Figure 6-58. VMware Workstation, GNS3 VM import in progress

06

Now, GNS3 VM has been created on your VMware Workstation 15 Pro and ready for use. You do not have to do anything with this VM at this stage as GNS3 software installation and integration are required for this VM to become useful. See Figure 6-59.

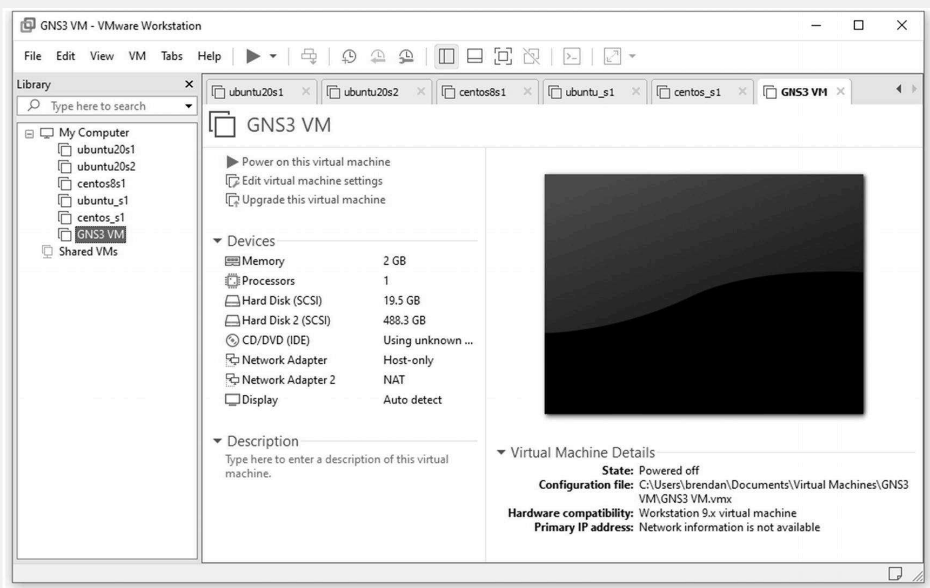


Figure 6-59. VMware Workstation, GNS3 VM installed

This completes downloading, importing, and installing the GNS3 VM for GNS3 integration later. Now that we have all the required VMs, we can learn about Linux basics, Python on Linux, GNS3, and Python network automation. Now the learning platform is almost complete.

Summary

You downloaded, installed, and created a CentOS 8 virtual machine in this chapter. You tested the SSH login to the newly created CentOS VM and learned an easy way to manage network adapter settings using the `nmtui` package method. Finally, you added the GNS3 VM in preparation for Cisco IOS integration later. Now you have two different flavors of Linux servers; you will learn the Linux basics in Chapter 7, followed by Linux software installation and administration in Chapter **8**.

Storytime 3: The Origin of Hypervisors

Earlier, you studied two different virtualization programs, and they are classified into either a Type-1 or Type-2 hypervisors. The hype around the cloud and cloud computing and on-demand IT concepts has been made because of the rapid development in hardware and software, specifically the virtualization technologies. If you look up the dictionary definition of *hypervisor*, you will soon discover that this word is a compound word made up of two words: *hyper* and *visor*. You can trace the origin to the Greek word *hyper*, meaning above, and the word *visor* is derived from the Old Anglo-French word *visor*, meaning to hide. **Wikipedia.com** suggests that *hypervisor* is a variant of the word *supervisor*. According to Wikipedia, the word *hypervisor* is the supervisor of supervisors. The word *hyper* carries a stronger synonymic meaning, at least linguistically, than the word *super*.

If you are also interested in DC Comics superhero characters, you know the most powerful character is Superman (although many will argue here). But if a Hyperman character was created and made famous through DC Comics, the Hyperman would be even stronger than Superman.

To understand the concept of virtualization more realistically, think of a hypervisor as the lubricant used in your car engine. A typical combustion engine car has larger components such as an engine and transmission. And inside a larger component, there are many smaller moving parts. Consider the hypervisor as the car engine's lubricant to prevent friction and wear on several smaller parts that reside inside the larger part. VM can run smoothly on its hosting server with very little or no compatibility issues. The hypervisor hides (or fabricates) the real hardware from the VM, thinking it is a real computer, but the VM is a handful of files running on hypervisors. The hypervisor is associated with the word *fabrication* in this sense, and the word *fabrication* means false. VMs think they are running on real hardware based servers, but all hardware are files, so the VMs are bunch of files running on a software.

Some of my clients' environments that I support still use the Cisco 6500/4500 Series of enterprise switches and Cisco 9000 Series switches with supervisor modules inserted into a large-framed switch body. If you are studying CCNA, you probably have heard of these enterprise-level switches. These switches can become a router, a switch, or both based on the OS and feature sets switched on. Some older engineers refer to these devices as SRouters because they can support multiple OSI layers. A supervisor in these devices is a modular hardware version of virtualization technology, running as a separate OS with its CPU and RAMs. Cisco had already begun hardware-based virtualization using various modules even before virtualization technologies took off years ago. For reference, the Nexus 3K/ 5K/6K/7K and Enterprise 9K switches offered on the current networking market support the Python APIs and come with Python 2.7.2 or newer by default. However, like many Cisco products, the Python version built into these switches has several restrictions and is limited to what you can do with it. For more information on these switches, visit **Cisco.com** (**https://www.cisco.com/c/en/us/products/switches/catalyst-9400-series-switches/index.html**).